

## **Compliance without Consensus: Global Investors, Board Gender Diversity, and External Scrutiny**

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### **Abstract**

As large institutional investors increasingly promote standardized governance practices across countries, convergence in governance practices does not necessarily imply convergence in how those practices are interpreted across institutional environments, particularly when global governance expectations intersect with distinct local institutional contexts. Using Japan's "Womenomics" reforms as a policy shift that elevated the salience of gender diversity within corporate governance, we show that firms with higher Big Three institutional ownership experience larger increases in board gender diversity following the reform, primarily through director replacement rather than board expansion. We further find that investor-induced increases in board gender diversity were associated with heightened governance-related scrutiny, reflected in higher levels of externally reported governance-related incidents, especially in male-dominated industries and among more visible firms. These increases are not accompanied by systematic declines in firm performance or governance quality indicators. Governance convergence may therefore generate organizational change without generating interpretive consensus.

**Keywords:** board gender diversity; governance convergence; global institutional investors; media scrutiny

## INTRODUCTION

Governance practices increasingly diffuse across institutional environments as large institutional investors promote similar governance expectations across firms and countries (Appel et al., 2016; Dyck et al., 2019). Prior research has largely seen this process as one of governance convergence, emphasizing how powerful investors encourage firms to adopt widely endorsed governance practices and diffuse governance expectations across firms and countries (e.g., Aggarwal et al., 2011; Azar et al., 2021; Dyck et al., 2019). However, adoption does not necessarily imply agreement regarding the desirability of those practices. When global and local actors hold divergent expectations or values, adopting globally promoted governance practices may generate tension rather than convergence because organizational responses remain embedded in local institutional environments (Schiehl et al., 2025; Witt et al., 2022; Wu et al., 2025). This raises a central question: when expectations diverge across institutional environments, how do firms adapt their governance practices in response to global pressures for conformity, and how are these changes interpreted within local contexts?

Prior research offers important insights into how firms respond to such tensions. Studies on institutional investors show that large asset managers increasingly promote standardized corporate governance practices across firms and countries, such as governance quality metrics (Aggarwal et al., 2011; Witt et al., 2022), ESG practices (Azar et al., 2021; Cohen, Kadach & Ormazabal, 2023; Dyck et al., 2019; Ilhan et al., 2023), and executive pay (Cohen, Kadach, Ormazaba & Reichelstein, 2023). At the same time, studies show that globally promoted practices may conflict with local governance arrangements because institutional environments differ in the expectations they place on firms (Aguilera & Jackson, 2003; Chung et al., 2024; Desender et al., 2016). Under these conditions, firms respond to conflicting governance

expectations through a range of strategies, including nonconformity (Witt et al., 2022), selective materiality (Wu et al., 2025), symbolic compliance (Ahmadjian & Robinson, 2001; Mun & Jung, 2018), and temporary compliance followed by reversal (Uchida, 2018).

Yet organizational responses do not necessarily resolve tensions between global governance standards and local institutional expectations. Governance changes remain subject to interpretation by external audiences, whose reactions can reinforce, reshape, or contest emerging governance norms (Phillips et al., 2004). Although prior research has examined how firms adapt to competing governance pressures, we know considerably less about how those governance changes are interpreted once implemented. This gap is important because the legitimacy of governance practices depends not only on their adoption, but also on how they are evaluated within the institutional environments in which firms operate (Bednar, 2012; Bednar et al., 2013). Domestic intermediaries, such as the media and other societal actors, play a particularly important role in this process because they interpret governance changes through country-specific institutional logics, cultural expectations, and understandings of appropriate corporate conduct. Consequently, governance practices promoted as desirable by global actors may not be interpreted in the same way within domestic contexts, particularly when different audiences evaluate those practices through distinct institutional expectations (Jones & Murray, 2026). Examining these interpretations is therefore essential for understanding when governance convergence produces shared governance expectations and when it generates contestation instead.

We examine this question in the context of gender diversity reforms in Japan. Japan provides a setting in which entrenched male-dominated norms coexist with growing global external demands for gender equality, while gender reform offers a policy context that amplifies

the existing tension between global and domestic actors. In 2013, the Japanese government introduced the “Womenomics” initiative, elevating gender diversity as a national policy while remaining largely non-binding in its implementation. Over the same decade, the “Big Three” institutional investors, including BlackRock, Vanguard, and State Street, actively promote gender diversity through their stewardship activities (Gormley et al., 2023). Leveraging the enhanced visibility and salience of gender diversity induced by the “Womenomics”, we examine how the Big Three affect firms’ adoption of board gender diversity, and how these changes are interpreted by domestic intermediaries, particularly domestic media and other societal actors (Bednar et al., 2013; Deephouse, 2000; Dyck et al., 2008; Kennedy, 2008).<sup>1</sup>

Our analysis proceeds in two stages. First, we show that firms with greater Big Three ownership experience larger increases in female board representation following the introduction of “Womenomics”. These changes typically occur through director replacement rather than board expansion, suggesting that firms more exposed to investor pressure adopt more organizationally consequential forms of adjustment. Second, we find that firms more exposed to investor-induced increases in board gender diversity experience higher levels of governance-related scrutiny, measured by governance-related risk incidents reported by the domestic media and other public sources. Importantly, the increase in reported governance-related incidents is not accompanied by a systematic decline in firm performance or governance quality indicators.

This study contributes to research on corporate governance, strategic leadership, and institutional environment by showing that governance convergence does not necessarily imply convergence in how governance changes are interpreted across institutional environments. In doing so, we highlight how macro-level institutional conditions shape not only the adoption of governance practices but also the meanings attached to those practices by domestic audiences.

Using a gender initiative as a lens, we show that transnational investor-driven changes can amplify firms' exposure to external scrutiny in contexts where local norms are misaligned with global standards. Together, we show that governance convergence can occur without interpretive convergence, i.e., firms may adopt governance practices promoted by global investors, yet those same practices may attract heightened scrutiny when evaluated by domestic audiences operating under different governance expectations.

## **GENDER REFORM, INVESTOR PRESSURE, AND SOCIAL INTERPRETATION IN JAPAN**

Japan provides a useful setting to examine how externally promoted governance reforms are implemented and interpreted domestically, particularly where support for gender diversity diverges from global expectations. Japan's corporate leadership has historically featured insider-dominated boards with few women (Ahmadjian & Robinson, 2001; Aoki, 2001). Its internal labor markets and seniority-based promotion systems historically limit women's progression into senior managerial roles (Brinton, 2001; Nemoto, 2013). As a result, expectations about who qualifies for board positions remain closely tied to male-dominated career trajectories, making the appointment of female directors more visible and subject to varying perceptions among local actors.

In 2013, the Japanese government introduced a set of policy initiatives commonly referred to as "Womenomics," under the administration of Prime Minister Shinzo Abe, aimed at increasing female participation in the workforce and in leadership roles (Abe, 2013). A central component of this initiative is encouraging firms to increase female representation in managerial and board positions (Japan Cabinet Office, 2013). The key difference from earlier initiatives is the introduction of concrete targets: raising the employment rate of women aged 25–44 to 73

percent and increasing women in leadership positions to 30 percent by 2020 (Government of Japan, 2014). These policies rely on voluntary compliance, and prior research suggests that such soft-law approaches increase attention to governance practices without mandating uniform responses (Mun & Jung, 2018). In this case, the initiative enhances board gender diversity as a prominent governance dimension, making it more visible and salient to society.

At the same time, elements of the reform increase the transparency of firms' gender-related practices. Disclosure requirements increase the accessibility of gender diversity metrics, enabling cross-firm comparison and making differences in board composition more salient to external observers. As a result, firms' responses to gender diversity expectations became more observable, comparable, and subject to interpretation by domestic external audiences.

These features of the reform are important for understanding firms' behavior in this setting. Because "Womenomics" relied on voluntary compliance, firms retained discretion over whether and how to adjust board composition, resulting in variation in both the likelihood and extent of change and allowing external pressure to shape outcomes. At the same time, increased transparency makes these choices more visible to societal actors. Thus, the reform does not mandate change but reshapes the conditions under which firms' governance decisions are observed and interpreted externally.

Over the same period, large global institutional investors, particularly BlackRock, Vanguard, and State Street, expand their ownership stakes in Japanese firms. These investors encourage firms to adopt governance practices aligned with global standards (Appel et al., 2016; Dyck et al., 2019), including the active promotion of board gender diversity as part of their stewardship activities (Gormley et al., 2023). Because ownership by these investors varies substantially across firms, their presence leads to variation in the external pressure faced by firms

to change gender diversity practices.

## **HYPOTHESES DEVELOPMENT**

We develop three sets of hypotheses examining whether firms respond to pressure from the Big Three, how they adjust to such pressure, and how resulting changes in board gender diversity are perceived and interpreted by external audiences.

### **Investor Pressure, Reform, and Board Gender Diversity**

Large institutional investors influence corporate governance by applying standardized expectations across firms in their portfolios, shaped by their domestic values and norms (Dyck et al., 2019). Through stewardship activities, these investors promote governance practices intended to enhance accountability and align firms with a single set of governance standards without tailoring to the institutional characteristics of individual firms or countries (Appel et al., 2016; Dyck et al., 2019; Azar et al., 2021; Gormley et al., 2023). For example, Dyck et al. (2019) show that institutional investors transplant their social norms regarding ESG issues around the world.

The Big Three institutional investors represent an influential subset of these investors due to the size and structure of their holdings (Aguilera et al., 2025). Their passive strategies further reinforce reliance on standardized governance policies rather than firm-specific monitoring (Appel et al., 2016; Azar et al., 2021).

Board gender diversity has become a central component of the Big Three's governance focus, following the growing emphasis on diversity, equity, and inclusion (DEI) in American society (Gormley et al., 2023). The Big Three have strong incentives to promote board gender diversity as part of their global stewardship agendas. Their size and visibility expose them to scrutiny from regulators, advocacy groups, and clients regarding their approach to responsible investment and governance (DesJardine et al., 2023; Edmans et al., 2019). Promoting gender

diversity allows the Big Three to signal alignment with US and global endorsed governance standards and to reinforce their reputations as responsible stewards of capital (Dyck et al., 2019; Kahan & Rock, 2020). In addition to these incentives, the Big Three possess the ability to translate stewardship priorities into firm-level pressure. Prior evidence indicates that the Big Three vote against director nominations at firms lacking female board representation and publicly communicate expectations regarding board diversity (Gormley et al., 2023). Consequently, differences in Big Three ownership translate into differences in firms' exposure to investor pressure to adjust board composition.

The effectiveness of the Big Three's pressure, however, depends on the policy context in which it is applied. The introduction of "Womenomics" elevates gender diversity as a policy priority in Japan, thereby enhancing the legitimacy of the Big Three's advocacy for such change in Japan (Suchman, 1995; Deephouse & Suchman, 2008). Moreover, this initiative increases the visibility of board gender diversity and strengthens disclosure of gender-related information. Under these conditions, firms with greater Big Three ownership become more exposed to governance expectations that could be more easily articulated, monitored, and enforced.

From the perspective of organizational responses to normative pressures, firms' awareness of gender equality and incentives to improve board gender diversity increase when such pressure comes from salient and influential investors backed by the "Womenomics" reform (Durand et al., 2019). The increased transparency and comparability linked to the reform, lower monitoring costs for these investors and raise the expected costs of non-compliance for more exposed firms. Accordingly, firms with greater Big Three ownership should be more likely to increase board gender diversity following the introduction of "Womenomics".

**Hypothesis 1 (H1):** *Big Three ownership is positively associated with board gender*



*diversity in Japan following the implementation of the “Womenomics” policy.*

### **Modes of Implementation: Board Restructuring versus Expansion**

Organizations vary not only in whether they respond to external pressures, but also in how they implement such responses. A central insight from Durand et al. (2019) is that organizational responses depend on both willingness to respond and the ability to implement change. When these conditions are only partially met, firms may adopt visible adjustments that signal alignment with external expectations without requiring substantial modification of existing structures (Meyer & Rowan, 1977; Oliver, 1991; Westphal & Zajac, 1994).

In the context of board gender diversity, firms can respond to external pressure through varying degrees of structural adjustment. According to Knippen et al. (2019), firms can increase female board representation either through board expansion that retains incumbents or through director substitution. Substitution is perceived as more substantial, as it requires firms to reallocate board seats and alter established board structures. It also signals a stronger commitment to gender diversity, because its implementation involves overcoming challenges arising from system inertia, particularly in Japan, where seniority-based promotion systems and stable director tenure make such changes difficult (Gilson & Roe, 1999; Mun & Jung, 2018).

We argue that exposure to the Big Three primarily affects firms’ likelihood to adjust board composition (Hypothesis 1). However, whether this likelihood translates into more structurally demanding changes depends on how the pressure interacts with the policy context, particularly the institutional context shaped by “Womenomics”. By increasing the visibility of board composition and introducing disclosure requirements, the reform reduces informational frictions for external observers by making board differences more observable and comparable across firms. As a result, compliance strategies also become more transparent. As expanding the board

while retaining incumbent directors may be perceived as a weaker form of compliance, firms facing stronger investor pressure therefore have incentives to substitute directors, which may be viewed as a more credible response by the Big Three.

Taken together, we expect the Big Three not only to increase the likelihood that firms respond to gender diversity pressure, but also to shape how these changes are implemented. Firms more exposed to the Big Three are therefore more likely to make changes that restructure existing board composition following the “Womenomics” reforms.

**Hypothesis 2 (H2):** *Big Three ownership is positively associated with increases in board gender diversity implemented through board restructuring rather than board expansion following the implementation of the “Womenomics” policy.*

### **Gender Diversity and Governance-related scrutiny**

Media and other societal actors, as public voice channels, play a central role in bringing firm behavior into the public domain. In particular, critical media coverage and societal actors’ views influence how firms are portrayed in the domestic context and which issues gain public attention (Deephouse, 2000; Bednar, 2012; Bednar et al., 2013; Eesley & Lenox, 2006). Given that these external actors have discretion over how they perceive and interpret firms, reported governance-related risk incidents may reflect either firms’ actual governance-related problems or external scrutiny in the absence of actual deterioration in governance.

Following “Womenomics,” board gender diversity becomes a more visible dimension of corporate governance, and a more salient object of external accounts, and we expect these effects to be stronger among firms with greater exposure to Big Three pressure. The implications of this visibility, whether positive or negative, depend on the institutional context in which such changes occur (Deephouse & Suchman, 2008; Suchman, 1995). Diversity initiatives are often promoted

globally as indicators of sound governance and organization progress (Kalev et al., 2006). However, governance practices are also interpreted through locally embedded norms and expectations (Winkler & Krzeminska, 2024). In Japan's historically male-dominated corporate environment, substantial increases in board gender diversity may constitute a visible deviation from established governance practices. Because such deviations can be interpreted both as governance improvements and as grounds for critical questioning, they become subject to competing interpretations regarding what constitutes appropriate governance and how organizational priorities are evolving (Castilla & Benard, 2010; Dobbin & Kalev, 2016; Eagly & Karau, 2002).

Domestic media and other societal actors play an important role in this process because they shape which governance issues receive public attention and how firms are evaluated (Bednar, 2012; Bednar et al., 2013; Deephouse, 2000). Visible departures from established governance practices are more likely to become subjects of public evaluation, particularly when their implications for corporate governance remain contested. For instance, domestic media and societal actors may question how changes in board composition affect existing governance arrangements, board dynamics, or organizational priorities. They may also interpret such changes as signaling a broader shift in the governance model of the firm, including a movement towards governance practices favored by global investors and greater emphasis on shareholder interests. As a result, board gender diversity may become a focal point through which broader debates about appropriate corporate governance are expressed. Such increased scrutiny does not require deterioration in underlying governance quality. We therefore expect firms more exposed to Big Three–induced increases in board gender diversity to face greater governance-related scrutiny from domestic media and other public sources following the introduction of “Womenomics”.

Accordingly, we propose:

**Hypothesis 3 (H3):** *In Japan, increases in board gender diversity are positively associated with governance-related scrutiny by domestic media and other societal actors.*

## DATA AND SUMMARY STATISTICS

### Empirical Strategy

We exploit the non-binding “Womenomics” reform as a policy-driven catalyst that raises the salience of gender diversity in Japan. The reform elevates board gender diversity as a visible and salient governance issue, thereby increasing its relevance for firms and external stakeholders. Rather than mandating changes in board composition, it gives firms discretion in their responses.

Our empirical strategy compares pre- and post-2013 changes in outcomes across firms with varying levels of ownership by the Big Three institutional investors. This design allows us to examine how exposure to global standardized governance expectations shapes Japanese firms’ responses to increased visibility of gender diversity. The estimated coefficients capture differential post-reform changes across firms with varying levels of Big Three ownership.

### Data and Sample

Our sample consists of Japanese listed firms over the 2006–2020 period. To construct a balanced pre- and post-treatment window, we focus on seven years before (2006–2012) and seven years after (2014–2020) the reform, treating 2013 as a transition year. This balanced panel facilitates a consistent comparison of pre- and post-reform dynamics.

We exclude financial firms because their regulatory environment and governance structures differ substantially from those of non-financial firms, which may differentially affect both ownership by global institutional investors and board composition decisions. Data on director characteristics, ownership structure, and financial variables are obtained from LSEG Refinitiv.

Governance characteristics, including board size and board independence, are hand-collected from firms' corporate governance reports filed with the Tokyo Stock Exchange.

We classify firms based on their exposure to Big Three ownership. For each industry-year, we calculate the mean Big Three ownership. Firms consistently at or above this mean from the pre- to post-reform period (2012–2020) are classified as the high-level group, while those consistently below form the low-level group. Firms that switch categories during this period are excluded. This design ensures that firms in the high- and low-level groups stay in their respective categories throughout the sample period. Defining treatment in 2012 ensures exogeneity to the reform. The final sample comprises 552 firms and 8,280 firm-year observations. Table I summarizes the sample construction process, including exclusions and the construction of the balanced panel.<sup>2</sup>

INSERT TABLE I ABOUT HERE

### **Key Variables**

**Board Gender Diversity:** We measure board gender diversity using two indicators. The first is the proportion of female directors on the board. The second is an indicator equal to one if the firm has at least one female director in a given year, and zero otherwise.

**Big Three Ownership:** Big Three ownership is measured as the percentage of shares held by the three largest global asset managers, BlackRock, Vanguard, and State Street Global Advisors. Consistent with prior research, we interpret this measure as capturing exposure to globally standardized stewardship practices and governance expectations.

**Governance-related scrutiny:** We measure governance-related scrutiny using governance-related incidents from RepRisk, a widely used database that aggregates reports from news media, NGOs, research firms, and other public sources. RepRisk compiles and codes governance-related

risk incidents based on publicly available information, classifying them into issue categories such as corruption, executive compensation, misleading communication, fraud, tax practices, and anti-competitive behavior. Importantly, RepRisk captures external reporting of governance-related risk incidents rather than verified deterioration in governance.

Although RepRisk does not provide the exact source for each incident, its “reach of information source” variable allows us to approximate whether the source is domestic or international. As shown in Online Appendix S1 Table AVII, for governance-related risk incidents, only 14.2 percent stem from global outlets, while the majority relate to local (23.8%) and regional/national resources (62.0%). Overall, over 85 percent of incidents are domestically sourced. Accordingly, RepRisk data provides a reasonable proxy for governance –related scrutiny that primarily reflects domestic actors.<sup>3</sup>

**Control Variables:** We include a standard set of firm-level controls that may be associated with both ownership structure and governance outcomes. These include ownership structure, firm size, index inclusion, board size, and board independence. Detailed definitions of all variables are provided in Online Appendix S1 Table AI.

### **Summary Statistics and Descriptive Evidence**

Table II reports summary statistics. On average, the Big Three hold 2.3 percent of shares across the full sample, with an average ownership of 7.8 percent among high-level firms and 0.6 percent among low-level firms. These ownership levels are comparable to those documented in prior studies examining the influence of large institutional investors (Gormley et al., 2023). Board gender diversity is relatively low compared to other countries with similar economic scales, with an average female ratio of 3.2 percent. Definitions of all variables are provided in Online Appendix S1, Table AI.

INSERT TABLE II ABOUT HERE

Figure 1 plots the average female director indicator and the proportion of female directors for firms with high versus low Big Three ownership. Consistent with the parallel trend assumption, board gender diversity follows similar trajectories across the two groups prior to 2013. After 2013, firms with higher Big Three ownership exhibit a faster increase in board gender diversity relative to firms in the low-level group.

INSERT FIGURE I ABOUT HERE

Prior to the reform, Big Three ownership varies substantially across firms. Consistent with prior research, firms with higher Big Three ownership tend to be larger, more likely to be included in global indices (e.g., MSCI and S&P 1200), and to have larger and more independent boards (Aggarwal et al., 2011; Gormley et al., 2023). Table III compares these characteristics between high- and low- level groups in 2012. Our empirical specification accounts for these differences using firm fixed effects and time-varying controls.

INSERT TABLE III ABOUT HERE

## EMPIRICAL ANALYSIS

### Specification

We estimate the following specification using the ordinary least squares model (OLS)<sup>4</sup>:

$$GenderDiv_{it+1} = \beta_1 Big3\ ownership_{it} \times Post2013_t + \beta_2 Big3\ ownership_{it} + \beta_3 X_{it} + \alpha_i + \delta_t + \epsilon_{it} \quad (1)$$

$GenderDiv_{it+1}$  denotes board gender diversity of firm  $i$  in year  $t + 1$ , measured alternatively as an indicator for the presence of at least one female director or as the proportion of female directors on the board. We use a one-year lead of the dependent variable to allow time for ownership structure and governance pressure to translate into board appointment decisions,

which typically occur on an annual cycle tied to shareholder meetings.

$Big3\ ownership_{it}$  measures the fraction of firm  $i$ 's equity held by the Big Three asset managers in year  $t$ .  $Post2013_t$  is an indicator equal to one for 2013 and all subsequent years and zero otherwise. The coefficient of interest,  $\beta_1$ , captures the differential change in board gender diversity after 2013 associated with greater Big Three ownership and constitutes the difference-in-differences estimator.

$X_{it}$  is a vector of time-varying ownership controls that capture the broader ownership structure of the firm and may influence board composition. These controls include *domestic ownership concentration*, *domestic bank ownership*, *domestic corporate ownership*, *domestic employee ownership*, *cross-shareholdings*, and *non-Big-Three foreign ownership*. Definitions of all variables are provided in Online Appendix S1 Table AI.

We include firm fixed effects ( $\alpha_i$ ) to account for time-invariant firm characteristics and year fixed effects ( $\delta_t$ ) to absorb common macroeconomic and regulatory shocks. Standard errors are clustered at the firm level to account for serial correlation in the error term.

Our identification relies on the assumption that, absent the reform, firms with differing levels of Big Three ownership would follow similar trends in board gender diversity. We examine the plausibility of this assumption using an event study specification.

## **Baseline Results**

Table IV reports the baseline estimates of the relationship between Big Three ownership and board gender diversity. Across specifications, firms with higher exposure to the Big Three exhibit larger increases in board gender diversity in the post-reform period.

In Panel A, Column (1), the interaction between Big Three ownership and the post-reform indicator is positive and statistically significant ( $\beta_1 = 0.486$ ,  $p < .001$ ), indicating that firms with



greater Big Three ownership experience larger increases in the proportion of female directors following the reform. The estimated effect in Column (11) is economically meaningful: a one-standard-deviation increase in Big Three ownership is associated with an increase in the female ratio on the board of approximately 0.012 ( $0.043 \times 0.29$ ), representing a 40 percent increase relative to the mean. Panel B reports consistent results using an indicator for the presence of at least one female director ( $\beta_1 = 2.206$ ,  $p < .001$ ).

We consider several alternative explanations. There may be confounding factors correlated with the Big Three ownership that can lead to different trends in outcomes. First, the observed effects may reflect broader changes in foreign institutional ownership rather than mechanisms specific to the Big Three. Column (2) includes an interaction between non-Big Three foreign ownership and the post-reform indicator to address this concern, and the results remain robust.

Second, firms' gender composition may be linked to Big Three ownership, and firms with no female directors may be more likely to add women over time (Gormley et al., 2023). Column (3) addresses this possibility by interacting the post-reform indicator with an indicator for firms with all-male boards in 2012. The results remain robust.

Third, variation in Big Three ownership is associated with firm characteristics such as size, index inclusion, and board structure. Columns (4)–(10) allow for differential post-reform trends along these dimensions by interacting the post indicator with pre-reform firm characteristics. The interaction effect with Big Three ownership remains stable across these specifications. Column (11) presents a specification including all dimensions simultaneously, with consistent results.

Overall, these findings provide evidence that firms with greater exposure to the Big Three experience larger increases in board gender diversity following the “Womenomics” reforms.<sup>5</sup>

INSERT TABLE IV ABOUT HERE

## Dynamic Effects and Parallel Trends

Table V and Figure 2 examine the timing of the effects. We estimate an event study specification that allows the effect of Big Three ownership to vary over time relative to the introduction of the “Womenomics” reforms.

Consistent with the parallel trends assumption, the estimates suggest that the interaction coefficients between Big Three ownership and year indicators for the pre-reform period (2006–2012) are small in magnitude and not statistically different from zero. The pre-treatment coefficients do not exhibit a systematic trend, providing no statistical evidence of differential pre-reform effects by Big Three ownership.

In contrast, the association between Big Three ownership and board gender diversity becomes positive and statistically significant in the post-reform period. The magnitude of the effect increases over time, indicating a sustained divergence between firms with higher and lower exposure to the Big Three.

Figure 2 illustrates that the post-reform increase in gender diversity is gradual rather than abrupt. This pattern is consistent with the time required for board turnover and director appointments, as well as with the incremental nature of governance change in the Japanese context (Mun and Jung, 2018). The slight and insignificant increase in 2011 and 2012 may be explained by smaller-scale gender-related efforts preceding “Womenomics”, including Japan’s 2011 commitment in the High-Level Policy Dialogue on Women and the Economy and the Tokyo Stock Exchange’s introduction of the “Nadeshiko Brand” in 2012. We address the potential pre-trend concerns related to earlier initiatives through the following points.

From the firms’ perspective, in theory, it is difficult for firms to anticipate the major 2013 reform. Early drafts and government discussions are vague and provide no clear requirements or

a predictable timeline. Moreover, gender diversity is a low-salience and low-priority governance issue in Japan before the reform, so early small-scale discussions do not trigger significant preparatory actions. In addition, before “Womenomics,” there are no strong precedents on gender diversity, leaving firms without a baseline to guide expectations or planning.

Empirically, even if firms fully anticipate the reform, such anticipation does not drive the estimated effects. Firms may act early to appoint women before the reform, which would mechanically attenuate our estimates. Nevertheless, we continue to observe statistically significant effects after 2013. Firms may also prepare for the reform without making early appointments. While this does not violate the parallel trends assumption, such preparation could correlate with firm characteristics, and thus with Big Three ownership, potentially creating differential post-reform trends. However, as shown in Table IV (Columns (4)–(10)), we control for these factors and find consistent results.

From the Big Three’s perspective, a potential concern is that they might anticipate and adjust their portfolios toward firms likely to increase female representation before 2013. By theory, these investors are unlikely to adjust their portfolios based primarily on female representation, regardless of other firm fundamentals. Even if the Big Three consider female representation as a basis for portfolio decisions, it is difficult for them to predict which firms are likely to increase the number of female directors, with 95 percent of firms in our sample having no female directors in 2012.

Moreover, we regress Big Three ownership on year indicators interacted with the female ratio in 2012. If anticipatory behaviors exist, we expect significant coefficients and clear trends before the reform. However, as shown in Online Appendix Table AII, the pre-2012 coefficients are small and statistically insignificant, with no clear pattern in their magnitude. We also fail to

reject the joint null that they are equal to zero ( $F = 0.38$ ,  $p = 0.8245$ ). These results provide no statistical evidence that the Big Three adjust investment before the reform, mitigating concerns that anticipatory portfolio adjustments drive our findings.

INSERT TABLE V AND FIGURE 2 ABOUT HERE

### **Modes of Implementation**

We next examine how firms increase female representation on their boards. Following Knippen et al. (2019), a more substantial change is captured by director substitution, defined as an increase in the number of female directors without a corresponding increase in board size. In contrast, board expansion is perceived as an easier move, defined as an increase in the number of female directors accompanied by an increase in board size.

Table VI reports the results. Firms with greater Big Three ownership are more likely to increase gender diversity through substitution rather than expansion following the reform. In Column (1), the interaction between Big Three ownership and the post-reform indicator is positive and statistically significant for substitution ( $\beta_1 = 0.354$ ,  $p < .001$ ), indicating that firms with higher Big Three ownership are more likely to replace male directors with female directors. The magnitude of the effect is economically meaningful relative to the baseline rate of substitution. In contrast, Column (2) shows no statistically significant effect on board expansion-based additions of female directors. A statistical test also supports that the effect on substitution is significantly larger than the effect on expansion ( $Z = 2.408$ ,  $p = 0.016$ ).

We complement our analysis by examining whether the Big Three increase firms' pursuit of third-party acknowledgment for their gender diversity practices, capturing a broader legitimacy-seeking response. To do so, we analyze firms' applications to and receipt of the "Nadeshiko Brand" designation, a government recognition awarded to firms demonstrating outstanding

practices in promoting women’s advancement. Columns (3) and (4) show that Big Three ownership is positively associated with both application to and receipt of the designation following the reform ( $\beta_1 = 3.308, p < .001$ ;  $\beta_1 = 0.860, p < .001$ ).

These results indicate that firms with higher Big Three ownership are more likely to change board gender composition in a more substantial manner and to receive public recognition of gender diversity practices.

INSERT TABLE VI ABOUT HERE

### **Gender Diversity and Governance-Related Scrutiny**

We next examine whether investor-induced increases in board gender diversity are associated with greater governance-related scrutiny. Because board gender diversity may be endogenous to firms' underlying governance characteristics, we proceed in three steps. We first document the baseline association between board gender diversity and governance-related incidents. We then estimate our primary empirical specification, which exploits the interaction between Big Three ownership and the introduction of the “Womenomics” reforms to isolate plausibly exogenous variation in board gender diversity. Finally, we estimate instrumental-variable (IV) models as an additional robustness check. We begin by estimating the following baseline specification:

$$Govern\ issues_{it+1} = \beta_1 GenderDiv_{it} + \beta_2 X_{it} + \alpha_i + \delta_t + \epsilon_{it} \quad (2)$$

where  $Govern\ issues_{it+1}$  is an indicator equal to one if firm  $i$  is associated with at least one governance-related incident in year  $t+1$ , and zero otherwise. All other variables are defined as in Equation (1).

Table VII reports the baseline results. Across all specifications, board gender diversity is positively associated with the likelihood of governance-related risk incidents ( $p < .001$ ). A one-

standard-deviation increase in the female director ratio and the female director indicator is associated with approximately a 32 percent and 30 percent increase, respectively, in the likelihood of reported governance-related incidents. These estimates establish the empirical association underlying Hypothesis 3 but should not be interpreted causally because firms with more gender-diverse boards may differ systematically from other firms in ways that also influence governance-related scrutiny.

INSERT TABLE VII ABOUT HERE

We therefore turn to our primary empirical specification, which exploits the differential increase in board gender diversity induced by the interaction between Big Three ownership and the post-reform period. As demonstrated in the preceding analyses, firms with greater Big Three ownership experienced significantly larger post-reform increases in female board representation, providing plausibly exogenous variation in board gender diversity. The reduced-form specification is given by:

$$\text{Govern issues}_{it+1} = \beta_1 \text{Big3 ownership}_{it} \times \text{Post2013}_t + \beta_2 \text{Big3 ownership}_{it} + \beta_3 X_{it} + \alpha_i + \delta_t + \epsilon_{it} \quad (3)$$

Table VIII reports the results for the reduced-form specification. In Column (1), the interaction between Big Three ownership and the post-reform indicator is positive and statistically significant ( $\beta_1 = 1.801$ ,  $p < .001$ ), indicating that firms with higher Big Three ownership experience an increase in externally reported governance-related incidents following the reform. The magnitude of the effect is economically meaningful relative to the baseline probability of reported incidents. This pattern is robust across alternative specifications that control for potential alternative channels, allowing for differential post-reform trends based on firm characteristics (Columns (2)–(10)).

These results are consistent with, but do not by themselves establish, the interpretation that firms more exposed to investor-driven gender diversity pressure have greater governance-related scrutiny. Importantly, this association does not necessarily indicate deteriorating governance quality, but rather reflects how firms are subject to domestic societal perception.

INSERT TABLE VIII ABOUT HERE

Finally, we estimate IV models that instrument board gender diversity using the interaction between Big Three ownership and the post-reform period as an additional robustness check. We present this robustness test in the Robustness Checks section.

Online Appendix S1, Table AVIII further examines the composition of governance-related incidents and shows that the association is not driven by a single category of reported issues. We interpret these results cautiously, as they reflect externally reported incidents rather than verified changes in underlying misconduct.

### **Additional Analyses: Interpreting Governance-Related Risk Incidents**

The observed rise in governance-related incidents could reflect either a genuine deterioration or increased scrutiny without actual deterioration. Although we cannot fully distinguish between these two scenarios, our analysis offers insights into which is more likely.

We first examine in which contexts these incidents are more concentrated. If incidents arise from external scrutiny without underlying deterioration, they should be more common where societal actors are more likely to notice and interpret divergent behavior critically. Highly visible firms and male-dominated industries are natural contexts to test this. More visible firms are easier targets for scrutiny and commentary. In male-dominated industries, gender diversity changes are more normatively salient and stand out more clearly against historical patterns, which increases the likelihood that they are subject to external debate. In contrast, if incidents

stem from actual deterioration behaviors, they should be higher where governance failure is more prevalent, regardless of visibility or industry gender composition.

Table IX reports results consistent with the absence of actual deterioration. Governance-related incidents are more likely to arise in male-dominated industries and firms with greater visibility, following the reform (Columns (1) – (5)). In contrast, we find no statistical evidence related to this argument among underperforming firms (Columns (6) – (8)).

INSERT TABLE IX ABOUT HERE

Second, if governance-related incidents reflect underlying deterioration, they should coincide with declines in firm performance, financial reporting quality, or external audit oversight. In contrast, if they reflect interpretive scrutiny, such incidents should not be associated with declines in those measures. As reported in Online Appendix S1, Tables AXIII and AXIV, we find no systematic evidence of declines in performance (e.g., financial performance, managerial outlook, and firm growth), or governance indicators related to financial reporting and external oversight (e.g., restatements, managerial forecast reporting quality, earnings surprises, and audit fees). In addition, these tests suggest that firm performance and financial reporting quality are unlikely to be the main channels driving the reduced-form results, leaving gender diversity as a plausible pathway.

Lastly, we examine whether the characteristics of reported governance incidents change following the reform. As shown in Online Appendix S1 Table AXV, the increase in reported incidents is not driven by greater severity or wider reach (i.e., coverage by more influential information resources), but rather by higher novelty. This pattern suggests that the rise in incidents reflects previously unreported issues that attract and sustain public attention, rather than a deterioration in realized governance conditions.



## Robustness Checks

### *Robustness checks: Big Three ownership, the reform, and board female representation*

First, to address concerns that Big Three ownership may respond to, rather than drive, firms' gender diversity initiatives, we re-estimate our models using pre-reform ownership measured in 2012 (Gormley et al., 2023). The results, reported in Online Appendix S1, Table AIII, remain consistent with our baseline findings. Similar results are obtained using a binary treatment definition based on pre-reform ownership (Online Appendix S1, Table AIV).

Second, we consider whether contemporaneous policy initiatives confound our estimates. In particular, the 2015 Act on Promotion of Women's Participation mandates firms with more than 300 employees to disclose gender-related metrics and action plans. Because firms subject to this requirement are likely to have higher Big Three ownership, observed increases in board gender diversity could reflect compliance with the Act rather than investor pressure. To address this concern, we allow for differential post-2015 trends among affected firms. As reported in Online Appendix S1 Table AV, the interaction between Big Three ownership and the post-reform period remains positive and statistically significant.

A further concern regarding the depth of gender diversity change relates to supply-side constraints in the market for female directors. Given the historically limited pool of women with prior board experience in Japan (Mun and Jung, 2018), firms facing diversity pressure may appoint the same individuals to multiple boards (Gormley et al., 2023). We examine whether increased Big Three ownership is associated with a rise in the number of board seats held by female directors after the reform and find no evidence of increased cross-board appointments among female directors (Online Appendix S1, Table AVI). This pattern suggests that firms expand the pool of female directors rather than concentrating appointments among a limited set

of individuals.

*Robustness checks: Gender diversity and governance-related scrutiny*

We estimate a two-stage least squares (2SLS) model in which female board representation is instrumented by the interaction between Big Three ownership and the post-reform period, and governance-related incidents are regressed on the instrumented female representation. This approach isolates the variation in female board representation that is driven by exposure to the “Womenomics” reform. The first-stage F-statistic exceeds the conventional threshold of 10, indicating the relevance of the instrument. The second stage results, reported in Online Appendix S1, Tables AXI and AXII, are consistent with our main findings.

Finally, we re-estimate the scrutiny models using pre-reform ownership measured in 2012. The results, reported in Online Appendix S1, Table AIX, remain consistent with the baseline findings, suggesting that the observed association is unlikely to be driven by post-reform adjustments in Big Three ownership.

## **DISCUSSION AND CONCLUSION**

This study examines how globally promoted governance practices are adopted and interpreted in institutional contexts where global governance expectations are not fully aligned with locally embedded governance norms. Focusing on the introduction of the “Womenomics” reforms in Japan, we show that firms with greater ownership by the Big Three institutional investors exhibit larger increases in board gender diversity following the reform. These changes are more likely to involve restructuring existing board composition rather than expanding board size. At the same time, firms more exposed to investor-induced increases in board gender diversity experience increases in externally reported governance-related incidents, particularly among more visible firms and in industries where gender diversity is less pronounced. These

findings show that global investor pressure and local institutional conditions jointly shape organizational change and how that change is subsequently evaluated by external audiences.

Our findings contribute to research on corporate governance and institutional environments by demonstrating that the effects of large institutional investors depend on the contexts in which their governance expectations are implemented. Prior work shows that the Big Three influence governance through standardized stewardship practices applied across diversified portfolios (Appel et al., 2016; Dyck et al., 2019; Azar et al., 2021; Gormley et al., 2023). We extend this literature by showing that institutional conditions influence not only whether investor pressure translates into organizational change, but also how governance changes are interpreted once implemented. While prior research has largely focused on the diffusion of governance practices, our findings suggest that the consequences of those practices depend on the institutional environments in which they are evaluated. Governance changes promoted by global investors may therefore generate different reactions across contexts even when firms adopt similar practices, extending recent research on contested governance reforms in East Asian institutional settings (Chung et al., 2024).

The study also advances research on organizational responses to institutional pressures by showing that external investor pressure can shape not only whether firms adopt governance practices, but also how governance structures are reconfigured in response. Firms with greater exposure to the Big Three are more likely to increase board gender diversity through director replacement rather than through board expansion. These findings suggest that sustained and standardized investor pressure influences not only whether firms respond to governance expectations, but also how governance changes are implemented, with more exposed firms adopting more consequential adjustments to existing governance structures. This complements

recent work on the governance consequences of gender diversity reforms (Duran et al., 2025).

More broadly, our findings contribute to research on organizational assessment by showing that governance practices may be evaluated differently across institutional environments, even when firms adopt similar governance changes. Governance practices promoted as desirable by global investors do not necessarily carry the same meaning across institutional environments, consistent with arguments that organizational practices acquire different meanings and consequences across contexts (Winkler & Krzeminska, 2024). In our setting, investor-induced increases in board gender diversity were associated with heightened governance-related scrutiny despite no systematic evidence of deteriorating governance quality. These findings suggest that governance convergence does not necessarily imply interpretive convergence, consistent with recent work emphasizing that legitimacy and evaluation may vary across institutional environments and audiences (Jones & Murray, 2026).

Taken together, these findings suggest that the consequences of governance changes cannot be understood solely in terms of adoption or adjustment. While global investors promote convergence toward widely endorsed governance standards, the reception of these practices depends on how they are interpreted in specific institutional contexts. This underscores that governance convergence may generate organizational change without generating convergence in how those changes are interpreted across audiences.

These insights also have implications for research on global governance and diversity and respond to recent calls for greater engagement with non-Western institutional contexts in management scholarship (Wickert et al., 2024). A growing literature examines how governance practices diffuse across firms and countries. Our results suggest that diffusion alone does not determine outcomes. While our setting is distinctive, the underlying mechanism, i.e.,

misalignment between globally promoted practices and local norms, may extend to other contexts where governance practices diffuse across heterogeneous institutional environments. The consequences of globally promoted governance practices depend on how they interact with local institutional environments. In contexts where gender diversity remains unevenly institutionalized, governance changes towards a global standard may increase visibility and scrutiny without generating uniformly favorable interpretations across audiences.

This study has several limitations that suggest directions for future research. First, our measure of externally reported governance-related incidents captures variation in publicly visible scrutiny, but does not allow us to disentangle underlying misconduct from variation in reporting intensity. Future research could build on this approach by examining how governance-related incidents are reported across different types of media and stakeholders. Second, our empirical setting focuses on Japan, where gender diversity has historically been limited, and reforms have been largely non-binding. While this context allows us to examine how governance changes are interpreted under conditions of incomplete alignment, future work could explore whether similar dynamics arise in settings with stronger regulatory mandates or more established norms. Third, while our findings are consistent with increased external attention and scrutiny following governance changes, further research could examine how specific intermediaries contribute to this process and how firms respond to heightened scrutiny over time.

In conclusion, this study shows that governance changes do more than alter organizational practices; they also shape the conditions under which those changes are perceived and interpreted. While global investors can promote substantial changes in corporate governance, these changes may also increase firms' exposure to governance-related scrutiny, particularly in contexts where audiences apply different criteria. Understanding this dual effect is critical for

assessing the broader implications of governance convergence across institutional environments.

## NOTES

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<sup>1</sup> We focus on board gender diversity for three reasons. First, it is a highly visible aspect of governance and often the first area affected by change (Mun & Jung, 2018). Second, boards are the primary decision-making and oversight bodies, so diversity at this level directly affects governance quality and strategic direction. Third, board composition is a key item in public reporting under the reform, providing reliable data and reducing measurement error and selection bias.

<sup>2</sup> Large language model-based artificial intelligence tools were used exclusively to assist with limited drafting and language editing of the manuscript text. No scientific content, data analysis, interpretation, or conclusions were generated by the tool. All content was reviewed and approved by the authors.

<sup>3</sup> We retain the full sample of incidents in all analyses, even though 14 percent of governance-related risk incidents come from high-reach global outlets. This is because some high-reach source incidents may be from the global media Japanese divisions. Excluding them may introduce bias or create the appearance of selective reporting consistent with our theory.

Nevertheless, our baseline results remain robust when the outcome variable is set to 1 only for limited- and medium-reach incidents. The results are shown in Online Appendix S1 Table AX.

<sup>4</sup> We use OLS regression, including when the dependent variable is binary, consistent with standard practice in difference-in-differences designs with panel data. In short panels, the fixed-effects Logit model is subject to the incidental parameters problem and yields inconsistent estimates. This concern is particularly salient in our setting, given the potential serial correlation

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of the dependent variable, which further exacerbates bias in nonlinear fixed-effects estimators. Hence, in this setting, OLS yields consistent estimates of average treatment effects under the parallel-trends and strict-exogeneity assumptions. OLS also facilitates straightforward adjustment of standard errors for within-firm serial correlation through clustering.

<sup>5</sup> The results are robust when the controls' interactions are specified with year dummies.

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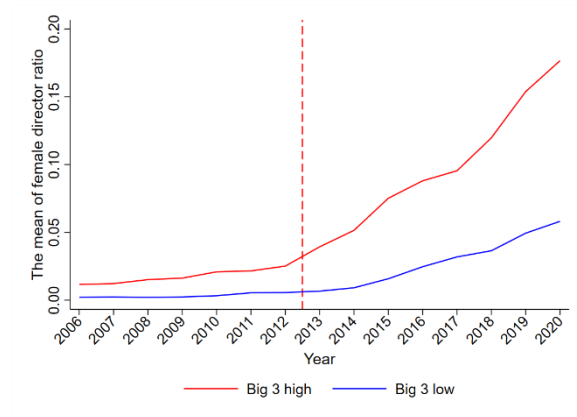
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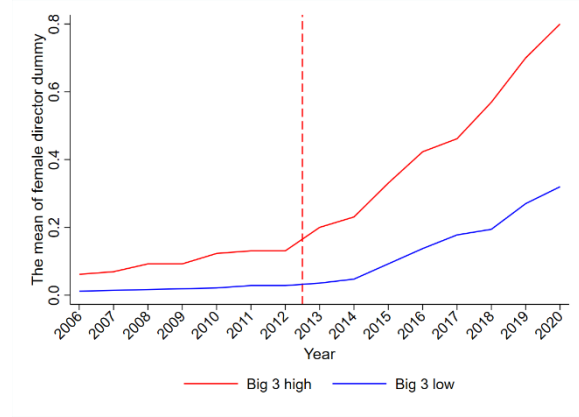
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**FIGURE 1** The trend of gender diversity on the board divided by the Big Three ownership level  
This figure plots the mean value of the female director ratio (Panel A) and the female director dummy (Panel B) on the board from 2006 to 2020, divided by the Big Three ownership level. Definitions of all variables are provided in Online Appendix S1 Table AI.

Panel A. The trend of female director ratio



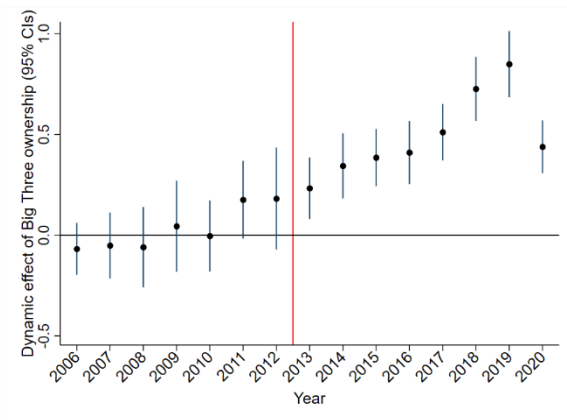
Panel B. The trend of female director dummy



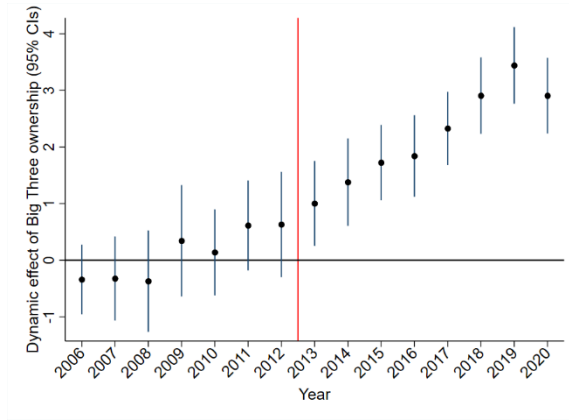
**FIGURE 2** Dynamic effects of Big Three shareholdings on gender diversity

This figure plots the marginal effects for each one-unit change in Big Three ownership on the female director ratio (Panel A) and the female director dummy (Panel B) on the board from 2006 to 2020. Vertical bars represent 95% confidence intervals. Robust standard errors are clustered at the firm level. Estimation by OLS regression. Definitions of all variables are provided in Online Appendix S1 Table AI.

Panel A. Dynamic effects on female director ratio



Panel B. Dynamic effects on female director dummy



**TABLE I** Sample construction

|                                                                                                                 |        |
|-----------------------------------------------------------------------------------------------------------------|--------|
| Observations with non-missing continuous values from 2006 to 2020                                               | 15570  |
| Observations in the financial industries                                                                        | (1641) |
| Observations with missing value of corporate governance variables, ownership variables and, financial variables | (804)  |
| Observations with firms switched between treated and control groups during 2012 - 2020                          | (4845) |
| Total firm-year observations                                                                                    | 8280   |



**TABLE II** Summary statistics

| Variable                              | N    | Mean   | SD    | Min    | Max    | p25    | p50    | p75    |
|---------------------------------------|------|--------|-------|--------|--------|--------|--------|--------|
| Female ratio                          | 8280 | 0.032  | 0.077 | 0.000  | 0.375  | 0.000  | 0.000  | 0.000  |
| Female dummy                          | 8280 | 0.175  | 0.380 | 0.000  | 1.000  | 0.000  | 0.000  | 0.000  |
| Female zero <sup>2012</sup>           | 8280 | 0.947  | 0.223 | 0.000  | 1.000  | 1.000  | 1.000  | 1.000  |
| Female director seats                 | 1269 | 1.399  | 1.518 | 0.000  | 6.000  | 0.000  | 1.000  | 2.000  |
| Substitute                            | 8280 | 0.038  | 0.190 | 0.000  | 1.000  | 0.000  | 0.000  | 0.000  |
| Add                                   | 8280 | 0.011  | 0.105 | 0.000  | 1.000  | 0.000  | 0.000  | 0.000  |
| Nadeshiko apply                       | 8280 | 0.095  | 0.294 | 0.000  | 1.000  | 0.000  | 0.000  | 0.000  |
| Nadeshiko win                         | 8280 | 0.018  | 0.134 | 0.000  | 1.000  | 0.000  | 0.000  | 0.000  |
| Big3 ownership                        | 8280 | 0.023  | 0.043 | 0.000  | 0.209  | 0.000  | 0.000  | 0.028  |
| Big3 ownership <sup>2012</sup>        | 8280 | 0.012  | 0.026 | 0.000  | 0.134  | 0.000  | 0.000  | 0.000  |
| Big3 ownership (high)                 | 1950 | 0.078  | 0.057 | 0.000  | 0.209  | 0.033  | 0.067  | 0.115  |
| Big3 ownership (low)                  | 6330 | 0.006  | 0.015 | 0.000  | 0.209  | 0.000  | 0.000  | 0.000  |
| Non-Big3 foreign ownership            | 8280 | 0.108  | 0.134 | 0.000  | 0.652  | 0.016  | 0.055  | 0.151  |
| Sales <sup>2012</sup>                 | 8280 | 20.907 | 1.538 | 17.987 | 24.623 | 19.770 | 20.675 | 21.906 |
| Market capitalization <sup>2012</sup> | 8280 | 19.938 | 1.697 | 17.091 | 24.212 | 18.670 | 19.527 | 21.202 |
| Assets <sup>2012</sup>                | 8280 | 20.972 | 1.551 | 18.095 | 24.946 | 19.846 | 20.619 | 21.976 |
| MSCI <sup>2012</sup>                  | 8280 | 0.042  | 0.200 | 0.000  | 1.000  | 0.000  | 0.000  | 0.000  |
| S&P1200 <sup>2012</sup>               | 8280 | 0.132  | 0.339 | 0.000  | 1.000  | 0.000  | 0.000  | 0.000  |
| Board independence <sup>2012</sup>    | 8280 | 0.113  | 0.142 | 0.000  | 0.556  | 0.000  | 0.071  | 0.182  |
| Board size <sup>2012</sup>            | 8280 | 8.734  | 3.156 | 4.000  | 19.000 | 6.000  | 8.000  | 10.000 |
| Domestic concentration                | 8280 | 0.171  | 0.158 | 0.014  | 0.800  | 0.075  | 0.115  | 0.203  |
| Domestic bank ownership               | 8280 | 0.134  | 0.122 | 0.000  | 0.484  | 0.025  | 0.109  | 0.212  |
| Domestic corporation ownership        | 8280 | 0.439  | 0.265 | 0.000  | 0.963  | 0.215  | 0.432  | 0.649  |
| Domestic employee ownership           | 8280 | 0.028  | 0.044 | 0.000  | 0.206  | 0.000  | 0.000  | 0.048  |
| Cross ownership                       | 8280 | 0.013  | 0.026 | 0.000  | 0.144  | 0.000  | 0.000  | 0.015  |

**TABLE III** Pre-Reform Comparison (2012)

|                                       | Sample average by Big3 ownership <sup>2012</sup> |                                    | p-value |
|---------------------------------------|--------------------------------------------------|------------------------------------|---------|
|                                       | Big3 ownership <sup>2012</sup> high              | Big3 ownership <sup>2012</sup> low |         |
| Market capitalization <sup>2012</sup> | 22.345                                           | 19.200                             | 0.00    |
| Sales <sup>2012</sup>                 | 22.934                                           | 20.279                             | 0.00    |
| Assets <sup>2012</sup>                | 23.149                                           | 20.301                             | 0.00    |
| MSCI <sup>2012</sup>                  | 0.146                                            | 0.009                              | 0.00    |
| S&P1200 <sup>2012</sup>               | 0.562                                            | 0.000                              | 0.00    |
| Board size <sup>2012</sup>            | 10.738                                           | 8.133                              | 0.00    |
| Board independence <sup>2012</sup>    | 0.205                                            | 0.085                              | 0.00    |
| N                                     | 130                                              | 422                                |         |

This table presents sample averages for the following firm characteristics in 2012 by Big Three ownership high and low groups based on mean value: Market capitalization, sales, assets, MSCI index inclusion, S&P1200 index inclusion, board size, and board independence. The first two columns present sample averages for firms in the high and low Big Three ownership, respectively. The last column provides the p-value from a t-test of the difference in means between the high and low groups. All variables are defined in Online Appendix S1 Table AI.

TABLE IV Baseline Estimates of Board Gender Diversity

| Panel A: Female ratio                            |                     |                     |                     |                      |                      |                     |                     |                     |                     |                     |                      |
|--------------------------------------------------|---------------------|---------------------|---------------------|----------------------|----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|----------------------|
|                                                  | (1)                 | (2)                 | (3)                 | (4)                  | (5)                  | (6)                 | (7)                 | (8)                 | (9)                 | (10)                | (11)                 |
| Big3 ownership                                   | 0.027<br>(0.076)    | 0.023<br>(0.077)    | 0.029<br>(0.077)    | 0.066<br>(0.075)     | 0.071<br>(0.075)     | 0.068<br>(0.075)    | 0.029<br>(0.073)    | 0.045<br>(0.082)    | 0.026<br>(0.076)    | 0.043<br>(0.075)    | 0.075<br>(0.072)     |
| Big3 ownership×Post 2013                         | 0.486***<br>(0.085) | 0.470***<br>(0.086) | 0.473***<br>(0.087) | 0.252**<br>(0.100)   | 0.274***<br>(0.093)  | 0.278***<br>(0.099) | 0.470***<br>(0.081) | 0.429***<br>(0.112) | 0.503***<br>(0.089) | 0.435***<br>(0.083) | 0.215**<br>(0.098)   |
| Domestic Concentration                           | 0.010<br>(0.020)    | 0.006<br>(0.019)    | 0.009<br>(0.019)    | 0.003<br>(0.019)     | 0.005<br>(0.019)     | 0.003<br>(0.019)    | 0.009<br>(0.019)    | 0.010<br>(0.020)    | 0.012<br>(0.020)    | 0.011<br>(0.019)    | 0.003<br>(0.018)     |
| Domestic bank ownership                          | 0.000<br>(0.019)    | -0.005<br>(0.020)   | 0.000<br>(0.019)    | -0.003<br>(0.019)    | 0.002<br>(0.019)     | 0.003<br>(0.019)    | 0.000<br>(0.019)    | 0.001<br>(0.019)    | -0.000<br>(0.019)   | -0.003<br>(0.019)   | -0.010<br>(0.019)    |
| Domestic corporate ownership                     | -0.008<br>(0.009)   | -0.010<br>(0.009)   | -0.007<br>(0.009)   | -0.004<br>(0.009)    | -0.002<br>(0.009)    | -0.001<br>(0.009)   | -0.007<br>(0.009)   | -0.007<br>(0.009)   | -0.009<br>(0.009)   | -0.009<br>(0.009)   | -0.007<br>(0.009)    |
| Domestic employee ownership                      | -0.018<br>(0.044)   | -0.023<br>(0.044)   | -0.018<br>(0.044)   | -0.010<br>(0.044)    | -0.018<br>(0.044)    | -0.016<br>(0.044)   | -0.016<br>(0.044)   | -0.016<br>(0.044)   | -0.017<br>(0.044)   | -0.024<br>(0.044)   | -0.013<br>(0.044)    |
| Cross ownership                                  | 0.017<br>(0.109)    | 0.012<br>(0.108)    | 0.016<br>(0.108)    | 0.008<br>(0.103)     | 0.013<br>(0.109)     | 0.005<br>(0.110)    | 0.006<br>(0.104)    | 0.020<br>(0.109)    | 0.017<br>(0.108)    | 0.030<br>(0.109)    | 0.006<br>(0.100)     |
| Non-Big3 foreign ownership                       | 0.010<br>(0.017)    |                     | 0.011<br>(0.017)    | 0.008<br>(0.017)     | 0.008<br>(0.017)     | 0.011<br>(0.017)    | 0.010<br>(0.017)    | 0.011<br>(0.017)    | 0.010<br>(0.017)    | 0.010<br>(0.017)    |                      |
| Non – Big3 ownership <sup>2012</sup> ×Post 2013  |                     | 0.011<br>(0.023)    |                     |                      |                      |                     |                     |                     |                     |                     | -0.014<br>(0.026)    |
| Female zero <sup>2012</sup> ×Post 2013           |                     |                     | -0.011<br>(0.017)   |                      |                      |                     |                     |                     |                     |                     | -0.003<br>(0.018)    |
| Market capitalization <sup>2012</sup> ×Post 2013 |                     |                     |                     | 0.009***<br>(0.002)  |                      |                     |                     |                     |                     |                     | 0.011***<br>(0.003)  |
| Sales <sup>2012</sup> ×Post 2013                 |                     |                     |                     |                      | 0.009***<br>(0.002)  |                     |                     |                     |                     |                     |                      |
| Assets <sup>2012</sup> ×Post 2013                |                     |                     |                     |                      |                      | 0.008***<br>(0.002) |                     |                     |                     |                     |                      |
| MSCI <sup>2012</sup> ×Post 2013                  |                     |                     |                     |                      |                      |                     | 0.019<br>(0.018)    |                     |                     |                     | 0.009<br>(0.018)     |
| S&P1200 <sup>2012</sup> ×Post 2013               |                     |                     |                     |                      |                      |                     |                     | 0.011<br>(0.013)    |                     |                     |                      |
| Board size <sup>2012</sup> ×Post 2013            |                     |                     |                     |                      |                      |                     |                     |                     | -0.001<br>(0.001)   |                     | -0.002***<br>(0.001) |
| Board independence <sup>2012</sup> ×Post 2013    |                     |                     |                     |                      |                      |                     |                     |                     |                     | 0.048**<br>(0.021)  | 0.034<br>(0.021)     |
| Constant                                         | 0.023***<br>(0.007) | 0.027***<br>(0.005) | 0.029**<br>(0.012)  | -0.068***<br>(0.024) | -0.079***<br>(0.026) | -0.071**<br>(0.028) | 0.023***<br>(0.007) | 0.023***<br>(0.007) | 0.027***<br>(0.008) | 0.022***<br>(0.007) | -0.074***<br>(0.028) |
| Firm FE                                          | Yes                 | Yes                 | Yes                 | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                  |
| Year FE                                          | Yes                 | Yes                 | Yes                 | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                  |
| N                                                | 8280                | 8280                | 8280                | 8280                 | 8280                 | 8280                | 8280                | 8280                | 8280                | 8280                | 8280                 |
| Adjusted R <sup>2</sup>                          | 0.560               | 0.560               | 0.560               | 0.564                | 0.564                | 0.563               | 0.560               | 0.560               | 0.560               | 0.562               | 0.567                |
| Panel B: Female dummy                            |                     |                     |                     |                      |                      |                     |                     |                     |                     |                     |                      |
|                                                  | (1)                 | (2)                 | (3)                 | (4)                  | (5)                  | (6)                 | (7)                 | (8)                 | (9)                 | (10)                | (11)                 |
| Big3 ownership                                   | 0.103<br>(0.294)    | 0.105<br>(0.291)    | 0.076<br>(0.293)    | 0.263<br>(0.293)     | 0.260<br>(0.295)     | 0.263<br>(0.298)    | 0.110<br>(0.289)    | 0.122<br>(0.310)    | 0.106<br>(0.294)    | 0.125<br>(0.294)    | 0.261<br>(0.289)     |
| Big3 ownership×Post 2013                         | 2.206***            | 2.123***            | 2.308***            | 1.234***             | 1.451***             | 1.399***            | 2.157***            | 2.145***            | 2.164***            | 2.135***            | 1.225***             |

|                                                  |          |          |         |          |          |          |          |          |          |          |          |
|--------------------------------------------------|----------|----------|---------|----------|----------|----------|----------|----------|----------|----------|----------|
|                                                  | (0.337)  | (0.350)  | (0.336) | (0.444)  | (0.417)  | (0.446)  | (0.337)  | (0.436)  | (0.353)  | (0.341)  | (0.438)  |
| Domestic Concentration                           | -0.021   | -0.032   | -0.008  | -0.050   | -0.041   | -0.049   | -0.025   | -0.021   | -0.025   | -0.020   | -0.036   |
|                                                  | (0.096)  | (0.090)  | (0.096) | (0.095)  | (0.095)  | (0.095)  | (0.096)  | (0.096)  | (0.096)  | (0.096)  | (0.090)  |
| Domestic bank ownership                          | -0.008   | -0.029   | -0.004  | -0.022   | 0.000    | 0.002    | -0.007   | -0.007   | -0.007   | -0.013   | -0.030   |
|                                                  | (0.095)  | (0.092)  | (0.094) | (0.095)  | (0.095)  | (0.095)  | (0.095)  | (0.095)  | (0.095)  | (0.095)  | (0.091)  |
| Domestic corporate ownership                     | -0.092*  | -0.098** | -0.099* | -0.076   | -0.070   | -0.067   | -0.090*  | -0.091*  | -0.090*  | -0.094*  | -0.088*  |
|                                                  | (0.051)  | (0.048)  | (0.051) | (0.051)  | (0.052)  | (0.052)  | (0.051)  | (0.051)  | (0.051)  | (0.051)  | (0.048)  |
| Domestic employee ownership                      | -0.060   | -0.078   | -0.078  | -0.029   | -0.060   | -0.053   | -0.054   | -0.058   | -0.062   | -0.069   | -0.051   |
|                                                  | (0.260)  | (0.259)  | (0.258) | (0.261)  | (0.261)  | (0.261)  | (0.258)  | (0.260)  | (0.259)  | (0.261)  | (0.261)  |
| Cross ownership                                  | 0.446    | 0.439    | 0.461   | 0.411    | 0.432    | 0.399    | 0.414    | 0.449    | 0.446    | 0.465    | 0.430    |
|                                                  | (0.587)  | (0.588)  | (0.602) | (0.555)  | (0.587)  | (0.584)  | (0.577)  | (0.587)  | (0.587)  | (0.587)  | (0.569)  |
| Non-Big3 foreign ownership                       | 0.028    |          | 0.021   | 0.020    | 0.022    | 0.031    | 0.028    | 0.030    | 0.028    | 0.028    |          |
|                                                  | (0.079)  |          | (0.079) | (0.079)  | (0.079)  | (0.079)  | (0.079)  | (0.079)  | (0.079)  | (0.079)  |          |
| Non – Big3 ownership <sup>2012</sup> ×Post 2013  |          | 0.078    |         |          |          |          |          |          |          |          | 0.006    |
|                                                  |          | (0.103)  |         |          |          |          |          |          |          |          | (0.110)  |
| Female zero <sup>2012</sup> ×Post 2013           |          |          | 0.108*  |          |          |          |          |          |          |          | 0.135**  |
|                                                  |          |          | (0.061) |          |          |          |          |          |          |          | (0.064)  |
| Market capitalization <sup>2012</sup> ×Post 2013 |          |          |         | 0.037*** |          |          |          |          |          |          | 0.042*** |
|                                                  |          |          |         | (0.011)  |          |          |          |          |          |          | (0.012)  |
| Sales <sup>2012</sup> ×Post 2013                 |          |          |         |          | 0.033*** |          |          |          |          |          |          |
|                                                  |          |          |         |          | (0.011)  |          |          |          |          |          |          |
| Assets <sup>2012</sup> ×Post 2013                |          |          |         |          |          | 0.033*** |          |          |          |          |          |
|                                                  |          |          |         |          |          | (0.012)  |          |          |          |          |          |
| MSCI <sup>2012</sup> ×Post 2013                  |          |          |         |          |          |          | 0.058    |          |          |          | 0.012    |
|                                                  |          |          |         |          |          |          | (0.073)  |          |          |          | (0.073)  |
| S&P1200 <sup>2012</sup> ×Post 2013               |          |          |         |          |          |          |          | 0.012    |          |          |          |
|                                                  |          |          |         |          |          |          |          | (0.051)  |          |          |          |
| Board size <sup>2012</sup> ×Post 2013            |          |          |         |          |          |          |          |          | 0.002    |          | -0.004   |
|                                                  |          |          |         |          |          |          |          |          | (0.004)  |          | (0.005)  |
| Board independence <sup>2012</sup> ×Post 2013    |          |          |         |          |          |          |          |          |          | 0.068    | 0.056    |
|                                                  |          |          |         |          |          |          |          |          |          | (0.091)  | (0.090)  |
| Constant                                         | 0.171*** | 0.180*** | 0.117** | -0.209*  | -0.193   | -0.194   | 0.170*** | 0.170*** | 0.162*** | 0.169*** | -0.312** |
|                                                  | (0.038)  | (0.027)  | (0.048) | (0.117)  | (0.128)  | (0.136)  | (0.038)  | (0.038)  | (0.041)  | (0.038)  | (0.129)  |
| Firm FE                                          | Yes      | Yes      | Yes     | Yes      | Yes      | Yes      | Yes      | Yes      | Yes      | Yes      | Yes      |
| Year FE                                          | Yes      | Yes      | Yes     | Yes      | Yes      | Yes      | Yes      | Yes      | Yes      | Yes      | Yes      |
| N                                                | 8280     | 8280     | 8280    | 8280     | 8280     | 8280     | 8280     | 8280     | 8280     | 8280     | 8280     |
| Adjusted R <sup>2</sup>                          | 0.551    | 0.551    | 0.551   | 0.554    | 0.553    | 0.553    | 0.551    | 0.551    | 0.551    | 0.551    | 0.555    |

This table reports coefficients from firm-panel regressions of female board representation on the interaction between a firm's Big Three ownership and an indicator for years in and after 2013. The dependent variables are the female ratio on the board (Panel A), and whether there are females on the board (Panel B). Column (2) includes a control for Non-Big Three foreign ownership in 2012 interacted with Post 2013. Column (3) includes a control for no female director indicator in 2012 interacted with Post2013. Column (4) includes a control for log market cap in 2012 interacted with Post2013. Column (5) includes a control for log assets in 2012 interacted with Post2013. Column (6) includes a control for log sales in 2012 interacted with Post2013. Column (7) includes a control for MSCI Index inclusion in 2012 interacted with Post2013. Column (8) includes a control for S&P 1200 Index inclusion in 2012 interacted with Post2013. Column (9) includes a control for board size in 2012 interacted with Post2013. Column (10) includes a control for board independence in 2012 interacted with Post2013. Column (11) includes controls for the Non-Big Three ownership, no-female-director indicator, log market capitalization, MSCI Index inclusion, board size, and board independence, with their 2012 values interacted with the Post-2013 indicator. The sample includes firm-year observations from 2006 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

**TABLE V** Dynamic Effects and Parallel Trends

| Dependent variable             | (1)<br>Female ratio | (2)<br>Female dummy |
|--------------------------------|---------------------|---------------------|
| Big3 ownership×Year = 2006     | -0.068<br>(0.066)   | -0.340<br>(0.313)   |
| Big3 ownership×Year = 2007     | -0.050<br>(0.083)   | -0.323<br>(0.378)   |
| Big3 ownership×Year = 2008     | -0.059<br>(0.102)   | -0.370<br>(0.456)   |
| Big3 ownership×Year = 2009     | 0.046<br>(0.115)    | 0.342<br>(0.501)    |
| Big3 ownership×Year = 2010     | -0.003<br>(0.090)   | 0.138<br>(0.387)    |
| Big3 ownership×Year = 2011     | 0.176*<br>(0.099)   | 0.613<br>(0.404)    |
| Big3 ownership×Year = 2012     | 0.181<br>(0.129)    | 0.630<br>(0.473)    |
| Big3 ownership×Year = 2013     | 0.233***<br>(0.078) | 1.002***<br>(0.382) |
| Big3 ownership×Year = 2014     | 0.344***<br>(0.082) | 1.379***<br>(0.393) |
| Big3 ownership×Year = 2015     | 0.385***<br>(0.072) | 1.724***<br>(0.338) |
| Big3 ownership×Year = 2016     | 0.410***<br>(0.080) | 1.841***<br>(0.367) |
| Big3 ownership×Year = 2017     | 0.512***<br>(0.071) | 2.329***<br>(0.329) |
| Big3 ownership×Year = 2018     | 0.727***<br>(0.081) | 2.908***<br>(0.343) |
| Big3 ownership×Year = 2019     | 0.850***<br>(0.084) | 3.443***<br>(0.345) |
| Big3 ownership×Year = 2020     | 0.462***<br>(0.068) | 2.882***<br>(0.339) |
| Constant                       | 0.020***<br>(0.007) | 0.148***<br>(0.037) |
| Controls                       | Yes                 | Yes                 |
| Firm FE                        | Yes                 | Yes                 |
| Year FE                        | Yes                 | Yes                 |
| <i>N</i>                       | 8280                | 8280                |
| Adjusted <i>R</i> <sup>2</sup> | 0.567               | 0.556               |

This table reports coefficients from firm-panel regressions of female board representation on the interaction between a firm's Big Three ownership and indicators for each year from 2006 through 2020. The dependent variables are the female ratio on the board (Column (1)), and whether there are females on the board (Column (2)). The sample includes firm-year observations from 2006 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

**TABLE VI** Degree of Gender Diversity Changes

| Dependent variable             | (1)<br>Substitution | (2)<br>Addition   | (3)<br>Nadeshiko apply | (4)<br>Nadeshiko win |
|--------------------------------|---------------------|-------------------|------------------------|----------------------|
| Big3 ownership                 | -0.001<br>(0.122)   | -0.003<br>(0.058) | -0.972***<br>(0.266)   | -0.293<br>(0.191)    |
| Big3 ownership×Post 2013       | 0.364***<br>(0.131) | 0.004<br>(0.060)  | 3.308***<br>(0.320)    | 0.860***<br>(0.187)  |
| Constant                       | 0.042***<br>(0.016) | 0.012<br>(0.008)  | 0.073***<br>(0.026)    | 0.007<br>(0.013)     |
| Z-statistic                    | 2.408               |                   |                        |                      |
| p-value                        | 0.016               |                   |                        |                      |
| Controls                       | Yes                 | Yes               | Yes                    | Yes                  |
| Firm FE                        | Yes                 | Yes               | Yes                    | Yes                  |
| Year FE                        | Yes                 | Yes               | Yes                    | Yes                  |
| <i>N</i>                       | 8280                | 8280              | 8280                   | 8280                 |
| Adjusted <i>R</i> <sup>2</sup> | 0.039               | 0.013             | 0.408                  | 0.247                |

This table reports coefficients from firm-panel regressions of substituting or adding female directors on the interaction between a firm's Big Three ownership and an indicator for years in and after 2013. The dependent variables are whether the females are appointed through substitution (Column (1)), whether females are appointed through extra addition (Column (2)), whether firms apply for the “Nadeshiko brand” (Column (3)), and whether firms are listed on the “Nadeshiko brand” (Column (4)). Z-statistic and p-value correspond to a z-test comparing the coefficients on the interaction terms in Columns (1) and (2). The sample includes firm-year observations from 2006 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

**TABLE VII** Governance-Related Risk Incidents (Baseline results)

|                                | (1)                 | (2)                  | (3)                 | (4)                  |
|--------------------------------|---------------------|----------------------|---------------------|----------------------|
| Female ratio                   | 0.370***<br>(0.080) | 0.336***<br>(0.077)  |                     |                      |
| Female dummy                   |                     |                      | 0.069***<br>(0.017) | 0.065***<br>(0.016)  |
| Market capitalization          |                     | 0.002<br>(0.010)     |                     | 0.001<br>(0.010)     |
| MSCI                           |                     | 0.025<br>(0.032)     |                     | 0.022<br>(0.033)     |
| Board size                     |                     | -0.004**<br>(0.002)  |                     | -0.005**<br>(0.002)  |
| Board independence             |                     | -0.043<br>(0.042)    |                     | -0.043<br>(0.042)    |
| Domestic Concentration         |                     | -0.023<br>(0.047)    |                     | -0.021<br>(0.047)    |
| Domestic bank ownership        |                     | -0.123*<br>(0.068)   |                     | -0.130*<br>(0.068)   |
| Domestic corporate ownership   |                     | -0.109***<br>(0.027) |                     | -0.109***<br>(0.027) |
| Domestic employee ownership    |                     | -0.288*<br>(0.149)   |                     | -0.292*<br>(0.151)   |
| Cross ownership                |                     | -0.600**<br>(0.260)  |                     | -0.638**<br>(0.260)  |
| Non-Big3 foreign ownership     |                     | -0.192***<br>(0.061) |                     | -0.197***<br>(0.061) |
| Constant                       | 0.067***<br>(0.002) | 0.178<br>(0.199)     | 0.068***<br>(0.002) | 0.201<br>(0.200)     |
| <i>N</i>                       | 8265                | 8265                 | 8265                | 8265                 |
| Adjusted <i>R</i> <sup>2</sup> | 0.387               | 0.392                | 0.386               | 0.391                |

This table reports coefficients from firm-panel regressions of governance risk incidents on the board female representation. The dependent variable is whether there are governance risk incidents. Columns (1) and (3) report specifications without control variables. Columns (2) and (4) include the full set of control variables. The sample includes firm-year observations from 2006 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

**TABLE VIII** Governance-Related Risk Incidents (Reduced form)

|                                                      | (1)                 | (2)                 | (3)                  | (4)                  | (5)                  | (6)                 | (7)                 | (8)                 | (9)                 | (10)                |
|------------------------------------------------------|---------------------|---------------------|----------------------|----------------------|----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Big3 ownership                                       | -0.209<br>(0.274)   | -0.200<br>(0.273)   | -0.101<br>(0.268)    | -0.037<br>(0.264)    | -0.023<br>(0.264)    | -0.202<br>(0.275)   | -0.016<br>(0.272)   | -0.206<br>(0.273)   | -0.167<br>(0.272)   | -0.070<br>(0.269)   |
| Big3 ownership×Post 2013                             | 1.801***<br>(0.253) | 1.770***<br>(0.255) | 1.143***<br>(0.301)  | 0.973***<br>(0.271)  | 0.860***<br>(0.288)  | 1.753***<br>(0.258) | 1.175***<br>(0.305) | 1.759***<br>(0.262) | 1.667***<br>(0.255) | 1.062***<br>(0.303) |
| Female zero <sup>2012</sup> ×Post 2013               |                     | -0.034<br>(0.028)   |                      |                      |                      |                     |                     |                     |                     | -0.012<br>(0.029)   |
| Market capitalization <sup>2012</sup> ×Post2013 2013 |                     |                     | 0.025***<br>(0.006)  |                      |                      |                     |                     |                     |                     | 0.024***<br>(0.007) |
| Sales <sup>2012</sup> ×Post 2013                     |                     |                     |                      | 0.036***<br>(0.006)  |                      |                     |                     |                     |                     |                     |
| Assets <sup>2012</sup> ×Post 2013                    |                     |                     |                      |                      | 0.038***<br>(0.007)  |                     |                     |                     |                     |                     |
| MSCI <sup>2012</sup> ×Post 2013                      |                     |                     |                      |                      |                      | 0.055<br>(0.053)    |                     |                     |                     | 0.034<br>(0.052)    |
| S&P1200 <sup>2012</sup> ×Post 2013                   |                     |                     |                      |                      |                      |                     | 0.124***<br>(0.034) |                     |                     |                     |
| Board size <sup>2012</sup> ×Post 2013                |                     |                     |                      |                      |                      |                     |                     | 0.002<br>(0.002)    |                     | -0.001<br>(0.002)   |
| Board independence <sup>2012</sup> ×Post 2013        |                     |                     |                      |                      |                      |                     |                     |                     | 0.128***<br>(0.044) | 0.096**<br>(0.043)  |
| Constant                                             | 0.080***<br>(0.022) | 0.097***<br>(0.024) | -0.177***<br>(0.066) | -0.319***<br>(0.069) | -0.345***<br>(0.077) | 0.079***<br>(0.022) | 0.071***<br>(0.022) | 0.071***<br>(0.023) | 0.076***<br>(0.022) | -0.157**<br>(0.073) |
| Controls                                             | Yes                 | Yes                 | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Firm FE                                              | Yes                 | Yes                 | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Year FE                                              | Yes                 | Yes                 | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| N                                                    | 8265                | 8265                | 8265                 | 8265                 | 8265                 | 8265                | 8265                | 8265                | 8265                | 8265                |
| Adjusted R <sup>2</sup>                              | 0.413               | 0.413               | 0.416                | 0.419                | 0.419                | 0.414               | 0.417               | 0.413               | 0.414               | 0.417               |

This table reports coefficients from firm-panel regressions of governance risk incidents on the interaction between a firm's Big Three ownership and an indicator for years in and after 2013. The dependent variable is whether there are governance risk incidents. Column (2) includes a control for no female director indicator in 2012 interacted with Post2013. Column (3) includes a control for log market cap in 2012 interacted with Post2013. Column (4) includes a control for log assets in 2012 interacted with Post2013. Column (5) includes a control for log sales in 2012 interacted with Post2013. Column (6) includes a control for MSCI Index inclusion in 2012 interacted with Post2013. Column (7) includes a control for S&P 1200 Index inclusion in 2012 interacted with Post2013. Column (8) includes a control for board size in 2012 interacted with Post2013. Column (9) includes a control for board independence in 2012 interacted with Post2013. Column (10) includes controls for the no-female-director indicator, log market capitalization, MSCI Index inclusion, board size, and board independence, with their 2012 values interacted with the Post-2013 indicator. The sample includes firm-year observations from 2006 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table A1.



**TABLE IX** Governance Scrutiny: Industry and Visibility Effects

|                                                                | (1)                         | (2)                   | (3)                   | (4)                  | (5)                 | (6)                 | (7)                 | (8)                 |
|----------------------------------------------------------------|-----------------------------|-----------------------|-----------------------|----------------------|---------------------|---------------------|---------------------|---------------------|
| Contingent factor                                              | No female director industry | Market capitalization | Sales                 | Assets               | Employee number     | ROE shortfall       | ROA shortfall       | Tobin's Q shortfall |
| Big3 ownership                                                 | -0.120<br>(0.264)           | 0.153<br>(2.505)      | -2.081<br>(3.076)     | -2.163<br>(3.084)    | -0.189<br>(1.078)   | -0.735**<br>(0.341) | -0.820**<br>(0.346) | -0.014<br>(0.445)   |
| Big3 ownership×Post 2013                                       | 1.598***<br>(0.244)         | -6.635**<br>(2.764)   | -13.287***<br>(3.273) | -9.981***<br>(3.219) | -3.043**<br>(1.209) | 1.902***<br>(0.317) | 1.921***<br>(0.331) | 1.783***<br>(0.350) |
| Big3 ownership×<br>Contingent factor <sup>2012</sup>           | -1.830<br>(1.248)           | -0.009<br>(0.126)     | 0.096<br>(0.147)      | 0.099<br>(0.146)     | 0.016<br>(0.138)    | 1.294**<br>(0.518)  | 1.398***<br>(0.510) | -0.398<br>(0.534)   |
| Post 2013× Contingent factor <sup>2012</sup>                   | -0.028<br>(0.017)           | 0.020***<br>(0.006)   | 0.024***<br>(0.006)   | 0.030***<br>(0.007)  | 0.024***<br>(0.006) | 0.004<br>(0.011)    | -0.001<br>(0.010)   | -0.002<br>(0.012)   |
| Big3 ownership×Post 2013×<br>Contingent factor <sup>2012</sup> | 2.759***<br>(0.985)         | 0.350***<br>(0.130)   | 0.619***<br>(0.150)   | 0.465***<br>(0.147)  | 0.437***<br>(0.142) | -0.288<br>(0.485)   | -0.345<br>(0.474)   | -0.079<br>(0.521)   |
| Constant                                                       | 0.083***<br>(0.022)         | -0.126*<br>(0.066)    | -0.198***<br>(0.070)  | -0.256***<br>(0.079) | -0.019<br>(0.033)   | 0.082***<br>(0.021) | 0.083***<br>(0.021) | 0.080***<br>(0.022) |
| Controls                                                       | Yes                         | Yes                   | Yes                   | Yes                  | Yes                 | Yes                 | Yes                 | Yes                 |
| Firm FE                                                        | Yes                         | Yes                   | Yes                   | Yes                  | Yes                 | Yes                 | Yes                 | Yes                 |
| Year FE                                                        | Yes                         | Yes                   | Yes                   | Yes                  | Yes                 | Yes                 | Yes                 | Yes                 |
| N                                                              | 8265                        | 8265                  | 8265                  | 8265                 | 7725                | 8265                | 8265                | 8265                |
| Adjusted R <sup>2</sup>                                        | 0.416                       | 0.418                 | 0.426                 | 0.423                | 0.430               | 0.416               | 0.416               | 0.414               |

This table reports coefficients from firm-panel regressions of governance risk incidents on interactions between a firm's Big Three ownership, an indicator for years in and after 2013, and a contingent factor. The contingent factors are measured using 2012 data, and include industry with no females (Column (1)), market capitalization (Column (2)), sales (Column (3)), assets (Column (4)), employee number (Column (5)), ROE shortfall (Column (6)), ROA shortfall (Column (7)), and Tobin's Q shortfall (Column (8)). The dependent variable is whether there are governance risk incidents. The sample includes firm-year observations from 2006 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

## **ONLINE APPENDIX S1. ADDITIONAL ANALYSES**

### **A. Overview of Online Appendix Structure**

This appendix provides additional analyses that support the identification strategy, robustness of the main findings, and interpretation of the results. The appendix is organized to parallel the structure of the main analysis, moving from identification and treatment validity to mechanism extensions, robustness and characterization of externally reported governance-related incidents, and tests that inform the interpretation of these incidents.

We begin by describing variable construction and data sources. Table AI provides detailed definitions of all variables used in the analysis, including dependent variables, key independent variables, and control variables.

We next present additional analyses that support the identification strategy underlying the difference-in-differences design. Figure A1 presents the marginal effects of Big Three ownership on governance-related risk incidents after reform, contingent on industry female composition, visibility, and underperformance. Table AII reports results by regressing Big Three ownership on year indicators interacted with the female ratio in 2012, addressing concerns that Big Three adjust their portfolios toward firms expected to increase female representation before 2013. Table AIII re-estimates the main results using pre-reform Big Three ownership measured in 2012, addressing concerns that ownership may respond endogenously to changes in board gender diversity. Table AIV reports results using a binary treatment definition based on whether firms' Big Three ownership exceeds the industry-year mean, demonstrating that the findings are not sensitive to the functional form of the ownership measure. Table AV examines robustness to the 2015 Act on Promotion of Women's Participation by allowing for differential trends among firms subject to disclosure requirements, ensuring that the estimated effects are not driven by contemporaneous policy changes.

We then provide additional evidence related to the mechanisms underlying changes in board composition. Table AVI examines whether firms respond to gender diversity pressure by appointing the same female directors across multiple boards, which would be consistent with supply constraints in the market for female directors. The results show no systematic increase in cross-board appointments, suggesting that firms expand the pool of female directors rather than concentrating appointments among a limited set of individuals.

The next set of analyses focuses on the robustness of externally reported governance-related incidents and their composition. Table AVII shows the distribution of governance-related incidents by the reach of the information source, providing evidence that the majority of the governance-related incidents have a domestic information source. Table AVIII examines the composition of governance-related incidents across issue categories, demonstrating that the results are not driven by any single type of incident. Table AIX re-estimates the relationship between Big Three ownership and governance-related incidents using pre-reform ownership, providing a robustness check analogous to Table AIII for the scrutiny outcome. Table AX re-estimates the relationship between Big Three ownership and governance-related incidents by redefining the outcome variable as 1 only when governance-related risk incidents have low- and medium- reach information sources.

We also provide additional analyses to isolate the variation in female board representation that is driven by exposure to the “Womenomics” reform. Tables AXI and AXII report instrumental variable estimates that use the interaction between Big Three ownership and the post-reform period as an instrument for board gender diversity.

Finally, we examine whether the observed increase in governance-related incidents reflects underlying deterioration in firm performance or reporting quality. Tables AXIII and AXIV

examine firm performance, financial reporting quality, and external audit oversight. Across these analyses, we find no systematic evidence that exposure to investor-driven gender diversity reforms is associated with declines in firm fundamentals, consistent with the interpretation that externally reported governance-related incidents reflect increased judgment rather than underlying deterioration. Table AXV examines whether the observed increase in incidents reflects changes in their characteristics, showing that the increase is not driven by greater severity or reach but is associated with greater novelty in reporting.

Finally, we provide the “Nadeshiko Brand” evaluation process derived from the Ministry of Economy, Trade and Industry website and regression models in robustness checks.

## B. Variables and Measurement

**TABLE AI** Variable definitions

| Variable                   | Definition                                                                                                                                                                                                                    | Source             |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Dependent variables</b> |                                                                                                                                                                                                                               |                    |
| Female ratio               | Female director number divided by board size                                                                                                                                                                                  | LSEG<br>Refinitive |
| Firm ratio <sup>2012</sup> | Female director number divided by board size in 2012                                                                                                                                                                          | LSEG<br>Refinitive |
| Female dummy               | An indicator variable that equals 1 if a firm has female directors on the board, and equals 0 in other cases.                                                                                                                 | LSEG<br>Refinitive |
| Substitution               | An indicator variable that equals 1 if in year $t + 1$ , there was an increase in the number of female directors but no increase in board size compared with year $t$ .                                                       | LSEG<br>Refinitive |
| Addition                   | An indicator variable that equals 1 if in year $t + 1$ , there was an increase in the number of female directors with an increase in board size compared with year $t$ and otherwise 0.                                       | LSEG<br>Refinitive |
| Nadeshiko apply            | An indicator variable that equals 1 if in year $t + 1$ , the firm applies for (with agreement to be disclosed) the “Nadeshiko Brand”, and otherwise 0.                                                                        | TSE website        |
| Nadeshiko win              | An indicator variable that equals 1 if in year $t + 1$ , the firm is listed on the “Nadeshiko Brand”, and otherwise 0.                                                                                                        | TSE website        |
| Female director seats      | Total seats a female director holds in other firms                                                                                                                                                                            | LSEG<br>Refinitive |
| Governance issues          | An indicator variable that equals 1 if a firm has governance risk incidents, including corruption, executive compensation, misleading communication, fraud, tax evasion, tax optimization, and anti-competition, otherwise 0. | Reprisk            |
| Severity indicator         | An indicator variable that equals 1 if a firm has the most severe (score = 3) governance risk incidents, otherwise 0.                                                                                                         | Reprisk            |

|                          |                                                                                                                                        |         |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------|
| Novelty indicator        | An indicator variable that equals 1 if a firm has the most novel (score = 2) governance risk incidents, otherwise 0.                   | Reprisk |
| Reach indicator          | An indicator variable that equals 1 if a firm has the most widely reached (score = 3) governance risk incidents, otherwise 0.          | Reprisk |
| Severity mean            | The year-mean of a firm's severity level of governance risk incidents.                                                                 | Reprisk |
| Novelty mean             | The year-mean of a firm's novelty level of governance risk incidents.                                                                  | Reprisk |
| Reach mean               | The year-mean of a firm's reach level of governance risk incidents.                                                                    | Reprisk |
| Severe×reach mean        | The year-mean of the multiplied severity and reach scores of a governance risk incident for a firm.                                    | Reprisk |
| Corruption               | An indicator variable that equals 1 if a firm has risk incidents on corruption, bribery, extortion, and money laundering, otherwise 0. | Reprisk |
| Executive compensation   | An indicator variable that equals 1 if a firm has risk incidents on executive compensation issues, otherwise 0.                        | Reprisk |
| Misleading communication | An indicator variable that equals 1 if a firm has risk incidents on misleading communication (including greenwashing), otherwise 0.    | Reprisk |
| Fraud                    | An indicator variable that equals 1 if a firm has risk incidents on fraud, otherwise 0.                                                | Reprisk |
| Tax evasion              | An indicator variable that equals 1 if a firm has risk incidents on tax evasion, otherwise 0.                                          | Reprisk |
| Tax optimization         | An indicator variable that equals 1 if a firm has risk incidents on tax optimization, otherwise 0.                                     | Reprisk |
| Anti competition         | An indicator variable that equals 1 if a firm has risk incidents on anti-competitive practices, otherwise 0.                           | Reprisk |

|                  |                                                                                                                                                         |                    |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| CEO optimism     | The first forecasting earning of the current year minus the previous year's realized earning, divided by last year's assets, scaled by one million.     | TSE website        |
| Revenue growth   | The increase in a company's total sales compared to the previous year.                                                                                  | LSEG<br>Refinitive |
| Employee growth  | The increase in a company's number of employees compared to the previous year.                                                                          | LSEG<br>Refinitive |
| Restatements     | An indicator variable if a firm has restatements.                                                                                                       | TSE website        |
| Revision number  | Total number a firm discloses earnings forecast revisions in one fiscal year.                                                                           | TSE website        |
| Forecast gap     | Absolute value of the first forecasting earning of the current year minus the realized earning, divided by previous year assets, scaled by one million. | TSE website        |
| Earning surprise | An indicator variable that equals 1 if the realized earning is smaller than the last forecasting earning, otherwise 0.                                  | TSE website        |
| Audit fee        | The natural logarithm of audit fees.                                                                                                                    | LSEG<br>Refinitive |

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#### **Independent variables**

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|                                              |                                                                                                       |                    |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------|
| Big3 ownership                               | The percentage of the total outstanding shares held by Blackrock, Vanguard, and State Street.         | LSEG<br>Refinitive |
| Big3 ownership <sup>2012</sup>               | The percentage of the total outstanding shares held by Blackrock, Vanguard, and State Street in 2012. | LSEG<br>Refinitive |
| Non-Big3 foreign ownership                   | The percentage of the total outstanding shares held by non-Big3 foreign investors.                    | LSEG<br>Refinitive |
| Non – Big3 foreign ownership <sup>2012</sup> | The percentage of the total outstanding shares held by non-Big3 foreign investors in 2012.            | LSEG<br>Refinitive |

|                                       |                                                                                                                                                                                                                                |                                       |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Big3 high                             | An indicator variable that equals 1 if a firm's Big Three holding is at or above year-industry mean level during 2012 – 2020, and equals 0 if a firm's Big Three holding is below year-industry mean level during 2012 – 2020. | LSEG<br>Refinitive                    |
| Post 2013                             | An indicator variable that equals 1 if a year is equal or larger than 2013, and 0 otherwise.                                                                                                                                   | LSEG<br>Refinitive                    |
| ROE                                   | The net income after tax over the total equity.                                                                                                                                                                                | LSEG<br>Refinitive                    |
| ROA                                   | The net income after tax over the total assets.                                                                                                                                                                                | LSEG<br>Refinitive                    |
| Employee 301                          | And indicator variable that equals 1 if the number of employees exceeds 300, and 0 otherwise.                                                                                                                                  | LSEG<br>Refinitive                    |
| Sales <sup>2012</sup>                 | The natural logarithm of sales in 2012.                                                                                                                                                                                        | LSEG<br>Refinitive                    |
| Market capitalization <sup>2012</sup> | The natural logarithm of market capitalization in 2012.                                                                                                                                                                        | LSEG<br>Refinitive                    |
| Assets <sup>2012</sup>                | The natural logarithm of book value of assets in 2012.                                                                                                                                                                         | LSEG<br>Refinitive                    |
| MSCI <sup>2012</sup>                  | An indicator variable if a firm is in the MSCI Global index in 2012                                                                                                                                                            | MSCI index<br>reconstitution<br>files |
| S&P1200 <sup>2012</sup>               | An indicator variable if a firm is in the S&P1200 Global index in 2012                                                                                                                                                         | WRDS                                  |
| Board size <sup>2012</sup>            | The number of directors on the board in 2012.                                                                                                                                                                                  | TSE website                           |
| Board independence <sup>2012</sup>    | The proportion of outside board members over the total board size in 2012.                                                                                                                                                     | TSE website                           |
| Domestic concentration                | The HHI index of domestic ownership.                                                                                                                                                                                           | LSEG<br>Refinitive                    |
| Domestic bank ownership               | The proportion of total outstanding shares held by domestic banks.                                                                                                                                                             | LSEG<br>Refinitive                    |



|                                     |                                                                                                               |                    |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------|
| Domestic corporate ownership        | The proportion of total outstanding shares held by domestic corporations.                                     | LSEG<br>Refinitive |
| Domestic employee ownership         | The proportion of total outstanding shares held by domestic employees.                                        | LSEG<br>Refinitive |
| Cross ownership                     | The sum of stakes held by other listed companies (where the focal firm also has a stake).                     | LSEG<br>Refinitive |
| No female industry <sup>2012</sup>  | An indicator variable that equals 1 if an industry has no female directors in 2012, otherwise 0.              | LSEG<br>Refinitive |
| Employee number <sup>2012</sup>     | The natural logarithm of employee numbers in 2012.                                                            | LSEG<br>Refinitive |
| ROE shortfall <sup>2012</sup>       | An indicator variable that equals 1 if a firm's ROE is above the industry average in 2012, otherwise 0.       | LSEG<br>Refinitive |
| ROA shortfall <sup>2012</sup>       | An indicator variable that equals 1 if a firm's ROA is above the industry average in 2012, otherwise 0.       | LSEG<br>Refinitive |
| Tobin's Q shortfall <sup>2012</sup> | An indicator variable that equals 1 if a firm's Tobin's Q is above the industry average in 2012, otherwise 0. | LSEG<br>Refinitive |

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## C. Figure for contingent effects

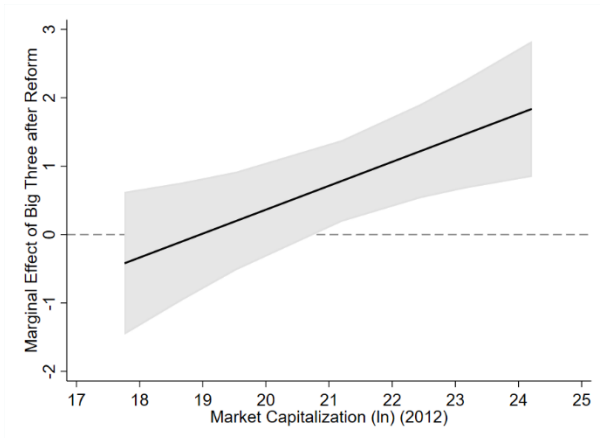
**FIGURE A1** Conditional Marginal Effects of Big Three after Reform

This figure plots the marginal effects for each one-unit change in Big Three ownership on governance-related risk incidents after reform, contingent on no female director (industry) dummy (Panel A1.1), market capitalization (Panel A1.2), sales (Panel A1.3), assets (Panel A1.4), employee number (Panel A1.5), ROE shortfall (Panel A1.6), ROA shortfall (Panel A1.7), and Tobin's Q shortfall (Panel A1.8). Vertical bars represent 95% confidence intervals. Robust standard errors are clustered at the firm level. Estimation by OLS regression. Definitions of all variables are provided in Online Appendix S1 Table AI.

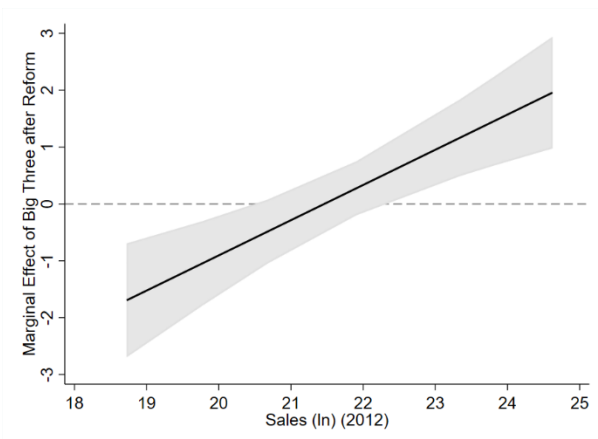
Panel A1.1 Contingent factor: No female director (ind.)



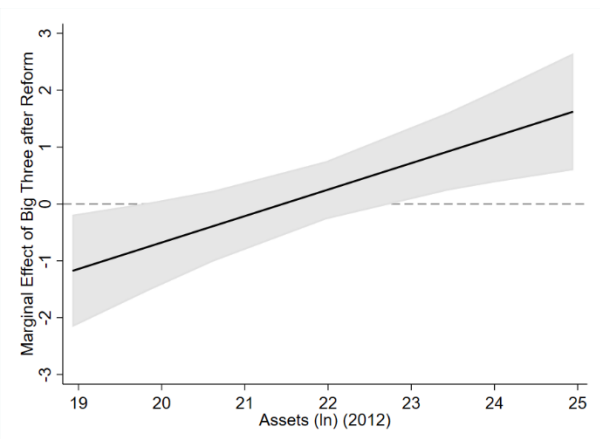
Panel A1.2 Contingent factor: Market capitalization



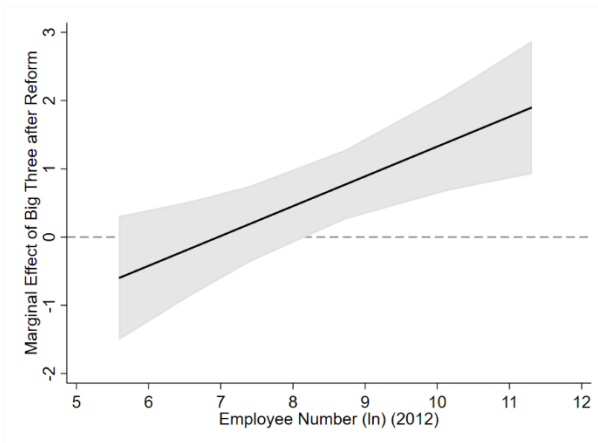
Panel A1.3 Contingent factor: Sales



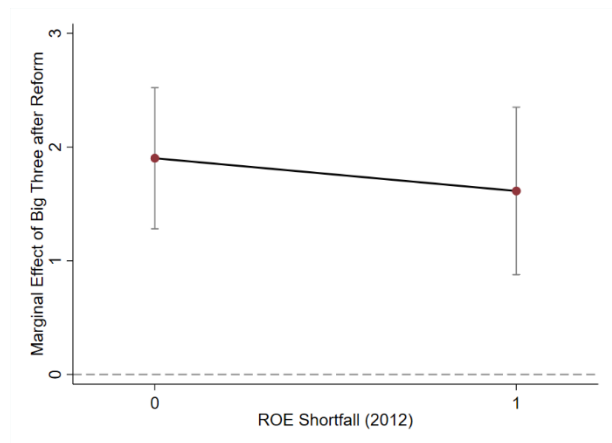
Panel A1.4 Contingent factor: Assets



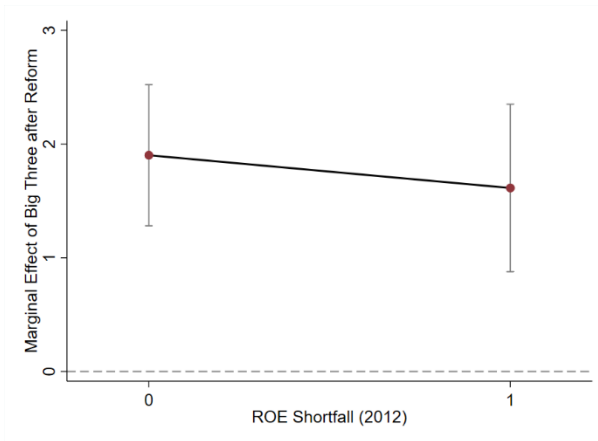
Panel A1.5 Contingent factor: Employee number



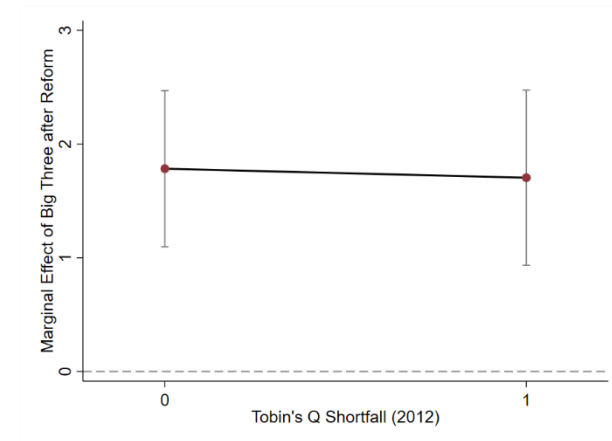
Panel A1.6 Contingent factor: ROE shortfall



Panel A1.7 Contingent factor: ROA shortfall



Panel A1.8 Contingent factor: Tobin's Q shortfall



## D. Identification and Treatment Validity

**TABLE AII** Pre-Reform Trends in Big Three Ownership: Evidence from Female Ratio (2012)

| Dependent variable                         | (1)<br>Big3 Ownership |
|--------------------------------------------|-----------------------|
| Female ratio <sup>2012</sup> × Year = 2008 | -0.015<br>(0.015)     |
| Female ratio <sup>2012</sup> × Year = 2009 | -0.020<br>(0.017)     |
| Female ratio <sup>2012</sup> × Year = 2010 | -0.010<br>(0.011)     |
| Female ratio <sup>2012</sup> × Year = 2011 | -0.001<br>(0.005)     |
| Female ratio <sup>2012</sup> × Year = 2013 | -0.004<br>(0.007)     |
| Female ratio <sup>2012</sup> × Year = 2014 | 0.015<br>(0.012)      |
| Female ratio <sup>2012</sup> × Year = 2015 | 0.038**<br>(0.019)    |
| Female ratio <sup>2012</sup> × Year = 2016 | 0.013<br>(0.018)      |
| Female ratio <sup>2012</sup> × Year = 2017 | 0.020<br>(0.017)      |
| Female ratio <sup>2012</sup> × Year = 2018 | 0.020<br>(0.017)      |
| Constant                                   | 0.056***<br>(0.003)   |
| Controls                                   | Yes                   |
| Firm FE                                    | Yes                   |
| Year FE                                    | Yes                   |
| <i>N</i>                                   | 21076                 |
| Adjusted R <sup>2</sup>                    | 0.663                 |

This table reports coefficients from regressions of Big Three Ownership on the interaction between a firm's female ratio on the board in 2012 and indicators for each year from 2008 through 2018. The dependent variable is the Big Three ownership. The sample includes firm-year observations from 2008 to 2018 and retains firms with non-continuous data as well as those that switch between high and low Big Three ownership groups. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

**TABLE AIII** Pre-Reform Big Three Ownership (2012) and Board Gender Diversity

| Dependent variable                            | (1)<br>Female<br>ratio | (2)<br>Female<br>dummy | (3)<br>Substitution | (4)<br>Addition  | (5)<br>Nadeshiko<br>apply | (6)<br>Nadeshiko<br>win |
|-----------------------------------------------|------------------------|------------------------|---------------------|------------------|---------------------------|-------------------------|
| Big3 ownership <sup>2012</sup> × Post<br>2013 | 0.660***<br>(0.137)    | 3.035***<br>(0.592)    | 0.617***<br>(0.185) | 0.045<br>(0.106) | 5.432***<br>(0.624)       | 1.246***<br>(0.292)     |
| Constant                                      | 0.044***<br>(0.008)    | 0.262***<br>(0.039)    | 0.053***<br>(0.016) | 0.011<br>(0.008) | 0.104***<br>(0.030)       | 0.017<br>(0.013)        |
| Z-statistic                                   | 2.540                  |                        |                     |                  |                           |                         |
| p-value                                       | 0.011                  |                        |                     |                  |                           |                         |
| Controls                                      | Yes                    | Yes                    | Yes                 | Yes              | Yes                       | Yes                     |
| Firm FE                                       | Yes                    | Yes                    | Yes                 | Yes              | Yes                       | Yes                     |
| Year FE                                       | Yes                    | Yes                    | Yes                 | Yes              | Yes                       | Yes                     |
| N                                             | 8280                   | 8280                   | 8280                | 8280             | 8280                      | 8280                    |
| Adjusted R <sup>2</sup>                       | 0.544                  | 0.538                  | 0.038               | 0.014            | 0.411                     | 0.245                   |

This table reports coefficients from firm-panel regressions of female board representation on the interaction between a firm's Big Three ownership in 2012 and an indicator for years in and after 2013. The dependent variables are the female ratio on the board (Column (1)), whether there are females on the board (Column (2)), whether the females are appointed through substitution (Column (3)), whether females are appointed through extra addition (Column (4)), whether firms apply for the "Nadeshiko brand" (Column (5)), and whether firms are listed on the "Nadeshiko brand" (Column (6)). Z-statistic and p-value correspond to a z-test comparing the coefficients on the interaction terms in Columns (3) and (4). The sample includes firm-year observations from 2006 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

**TABLE AIV** Binary Independent Variable Specification: Big Three Ownership and Board Gender Diversity

|                           | (1)<br>Female<br>ratio | (2)<br>Female<br>dummy | (3)<br>Substitution | (4)<br>Addition  | (5)<br>Nadeshiko<br>apply | (6)<br>Nadeshiko<br>win |
|---------------------------|------------------------|------------------------|---------------------|------------------|---------------------------|-------------------------|
| Big3<br>high×Post<br>2013 | 0.054***<br>(0.008)    | 0.229***<br>(0.033)    | 0.042***<br>(0.011) | 0.005<br>(0.006) | 0.363***<br>(0.033)       | 0.091***<br>(0.017)     |
| Constant                  | 0.035***<br>(0.008)    | 0.229***<br>(0.038)    | 0.049***<br>(0.016) | 0.010<br>(0.008) | 0.067***<br>(0.026)       | 0.005<br>(0.014)        |
| Z-statistic               | 2.866                  |                        |                     |                  |                           |                         |
| p-value                   | 0.004                  |                        |                     |                  |                           |                         |
| Controls                  | Yes                    | Yes                    | Yes                 | Yes              | Yes                       | Yes                     |
| Firm FE                   | Yes                    | Yes                    | Yes                 | Yes              | Yes                       | Yes                     |
| Year FE                   | Yes                    | Yes                    | Yes                 | Yes              | Yes                       | Yes                     |
| N                         | 8280                   | 8280                   | 8280                | 8280             | 8280                      | 8280                    |
| Adjusted R <sup>2</sup>   | 0.553                  | 0.543                  | 0.038               | 0.014            | 0.419                     | 0.250                   |

This table reports coefficients from firm-panel regressions of female board representation on the interaction between a firm's Big Three ownership level, measured using dummies and an indicator for years in and after 2013. The dependent variables are the female ratio on the board (Column (1)), whether there are females on the board (Column (2)), whether the females are appointed through substitution (Column (3)), whether females are appointed through extra addition (Column (4)), whether firms apply for the "Nadeshiko brand" (Column (5)), and whether firms are listed on the "Nadeshiko brand" (Column (6)). Z-statistic and p-value correspond to a z-test comparing the coefficients on the interaction terms in Columns (3) and (4). The sample includes firm-year observations from 2006 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Appendix S1 Table AI.

**TABLE AV** Robustness to the 2015 Gender Disclosure Policy

| Dependent variable       | (1)<br>Female ratio | (2)<br>Female dummy | (3)<br>Substitution | (4)<br>Addition    | (5)<br>Nadeshiko apply | (6)<br>Nadeshiko win |
|--------------------------|---------------------|---------------------|---------------------|--------------------|------------------------|----------------------|
| Big3 ownership           | 0.025<br>(0.077)    | 0.080<br>(0.296)    | -0.006<br>(0.120)   | 0.005<br>(0.058)   | -1.002***<br>(0.268)   | -0.307<br>(0.201)    |
| Big3 ownership×Post 2013 | 0.485***<br>(0.085) | 2.224***<br>(0.338) | 0.367***<br>(0.122) | 0.001<br>(0.061)   | 3.296***<br>(0.323)    | 0.876***<br>(0.197)  |
| Employee 301×Year = 2006 | 0.010<br>(0.011)    | 0.111*<br>(0.065)   | 0.010<br>(0.018)    | 0.005<br>(0.013)   | 0.007<br>(0.031)       | 0.018<br>(0.017)     |
| Employee 301×Year = 2007 | 0.011<br>(0.011)    | 0.122*<br>(0.071)   | 0.006<br>(0.019)    | 0.011<br>(0.013)   | 0.016<br>(0.039)       | 0.027<br>(0.022)     |
| Employee 301×Year = 2008 | 0.008<br>(0.010)    | 0.117<br>(0.074)    | 0.002<br>(0.020)    | 0.001<br>(0.011)   | 0.008<br>(0.036)       | 0.019<br>(0.019)     |
| Employee 301×Year = 2009 | 0.010<br>(0.010)    | 0.124*<br>(0.073)   | 0.001<br>(0.020)    | 0.008<br>(0.012)   | 0.001<br>(0.035)       | 0.018<br>(0.018)     |
| Employee 301×Year = 2010 | 0.011<br>(0.010)    | 0.133*<br>(0.072)   | 0.003<br>(0.020)    | 0.002<br>(0.011)   | 0.004<br>(0.036)       | 0.020<br>(0.019)     |
| Employee 301×Year = 2011 | 0.006<br>(0.009)    | 0.076<br>(0.060)    | -0.003<br>(0.018)   | -0.024<br>(0.032)  | 0.019<br>(0.033)       | 0.031*<br>(0.017)    |
| Employee 301×Year = 2012 | 0.016<br>(0.010)    | 0.129*<br>(0.068)   | 0.012<br>(0.021)    | 0.009<br>(0.014)   | 0.015<br>(0.028)       | 0.038**<br>(0.015)   |
| Employee 301×Year = 2013 | 0.010<br>(0.010)    | 0.096<br>(0.067)    | -0.001<br>(0.021)   | 0.009<br>(0.014)   | 0.072*<br>(0.037)      | 0.030**<br>(0.015)   |
| Employee 301×Year = 2014 | 0.017<br>(0.010)    | 0.112<br>(0.071)    | 0.017<br>(0.022)    | 0.007<br>(0.037)   | 0.131***<br>(0.032)    | 0.032*<br>(0.016)    |
| Employee 301×Year = 2015 | 0.017*<br>(0.010)   | 0.104<br>(0.074)    | 0.009<br>(0.043)    | -0.002<br>(0.012)  | 0.107***<br>(0.035)    | 0.030*<br>(0.017)    |
| Employee 301×Year = 2016 | 0.017<br>(0.012)    | 0.151**<br>(0.077)  | -0.008<br>(0.046)   | 0.008<br>(0.014)   | 0.120***<br>(0.035)    | 0.026<br>(0.017)     |
| Employee 301×Year = 2017 | 0.024**<br>(0.012)  | 0.174**<br>(0.077)  | 0.041*<br>(0.023)   | 0.010<br>(0.013)   | 0.033<br>(0.035)       | 0.016<br>(0.015)     |
| Employee 301×Year = 2018 | 0.036***<br>(0.011) | 0.195***<br>(0.075) | 0.055<br>(0.047)    | -0.031<br>(0.042)  | 0.075**<br>(0.032)     | 0.022<br>(0.015)     |
| Employee 301×Year = 2019 | 0.032**<br>(0.013)  | 0.168**<br>(0.084)  | -0.025<br>(0.061)   | -0.018<br>(0.036)  | 0.034<br>(0.038)       | 0.022<br>(0.015)     |
| Employee 301×Year = 2020 | 0.001<br>(0.015)    | 0.055<br>(0.108)    | -0.092<br>(0.082)   | 0.027**<br>(0.013) | 0.041<br>(0.040)       | 0.020<br>(0.014)     |
| Constant                 | 0.007<br>(0.011)    | 0.044<br>(0.067)    | 0.042*<br>(0.024)   | 0.009<br>(0.015)   | 0.026<br>(0.039)       | -0.018<br>(0.020)    |
| Z-statistic              |                     |                     | 2.408               |                    |                        |                      |
| p-value                  |                     |                     | 0.016               |                    |                        |                      |
| Controls                 | Yes                 | Yes                 | Yes                 | Yes                | Yes                    | Yes                  |
| Firm FE                  | Yes                 | Yes                 | Yes                 | Yes                | Yes                    | Yes                  |
| Year FE                  | Yes                 | Yes                 | Yes                 | Yes                | Yes                    | Yes                  |
| N                        | 8064                | 8064                | 8064                | 8064               | 8064                   | 8064                 |
| Adjusted R <sup>2</sup>  | 0.557               | 0.549               | 0.039               | 0.014              | 0.401                  | 0.246                |

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This table reports coefficients from firm-panel regressions of female board representation on the interaction between a firm's Big Three ownership and an indicator for years in and after 2013, and interactions between an indicator for having more than 300 employees and indicators for each year from 2006 through 2020. The dependent variables are the female ratio on the board (Column (1)), whether there are females on the board (Column (2)), whether the females are appointed through substitution (Column (3)), whether females are appointed through extra addition (Column (4)), whether firms apply for the "Nadeshiko brand" (Column (5)), and whether firms are listed on the "Nadeshiko brand" (Column (6)). Z-statistic and p-value correspond to a z-test comparing the coefficients on the interaction terms in Columns (3) and (4). The sample includes firm-year observations from 2006 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.



## E. Mechanism Extension: Director Supply and Board Appointments

**TABLE AVI** Female Director Cross-Board Appointments

| Dependent variable             | (1)<br>Female director seats |
|--------------------------------|------------------------------|
| Big3 ownership                 | -5.421<br>(3.996)            |
| Big3 ownership×Post 2013       | 5.835<br>(4.106)             |
| Constant                       | 1.392***<br>(0.062)          |
| Director FE                    | Yes                          |
| Year FE                        | Yes                          |
| <i>N</i>                       | 1269                         |
| Adjusted <i>R</i> <sup>2</sup> | 0.770                        |

This table reports coefficients from director-panel regressions of female board seat numbers on the interaction between a firm's Big Three ownership and an indicator for years in and after 2013. The dependent variable is the number of seats female directors hold in other firms. The sample includes director-year observations from 2006 through 2020. Standard errors, which are adjusted for clustering at the director level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

## F. External Scrutiny: Source, Composition, and Robustness

**TABLE AVII** Distribution of Governance-related Risk Incident by Reach (2007–2020)

| Reach level | Description   | Source                                                                           | Percent (%) |
|-------------|---------------|----------------------------------------------------------------------------------|-------------|
| 1           | Limited reach | Local media, local governmental bodies, smaller NGOs, and social media           | 23.80       |
| 2           | Medium reach  | Most national/regional media, state/regional governments, and international NGOs | 62.01       |
| 3           | High reach    | Global outlets                                                                   | 14.19       |

This table shows the distribution of governance-related risk incidents by the reach of the information source. The data are incident-based and cover the period from 2007 to 2020. Full definitions of source and descriptions for each category are available on the RepRisk website (Methodology section).

**TABLE AVIII** Composition of Governance-Related Risk Incidents

| Dependent variable             | (1)<br>Corruption   | (2)<br>Executive<br>compensation | (3)<br>Misleading<br>communication | (4)<br>Fraud        | (5)<br>Tax<br>evasion | (6)<br>Tax<br>optimization | (7)<br>Anti<br>competition |
|--------------------------------|---------------------|----------------------------------|------------------------------------|---------------------|-----------------------|----------------------------|----------------------------|
| Big3 ownership                 | 0.036<br>(0.214)    | -0.026<br>(0.020)                | -0.060<br>(0.081)                  | -0.181<br>(0.217)   | -0.016<br>(0.041)     | 0.002<br>(0.032)           | -0.043<br>(0.244)          |
| Big3 ownership×Post 2013       | 0.471***<br>(0.178) | 0.015<br>(0.024)                 | 0.605***<br>(0.130)                | 1.163***<br>(0.224) | 0.157**<br>(0.066)    | 0.100***<br>(0.037)        | 0.993***<br>(0.263)        |
| Constant                       | 0.024*<br>(0.013)   | 0.006**<br>(0.003)               | 0.016<br>(0.010)                   | 0.022<br>(0.015)    | -0.003<br>(0.005)     | -0.006*<br>(0.003)         | 0.029<br>(0.019)           |
| Controls                       | Yes                 | Yes                              | Yes                                | Yes                 | Yes                   | Yes                        | Yes                        |
| Firm FE                        | Yes                 | Yes                              | Yes                                | Yes                 | Yes                   | Yes                        | Yes                        |
| Year FE                        | Yes                 | Yes                              | Yes                                | Yes                 | Yes                   | Yes                        | Yes                        |
| <i>N</i>                       | 8265                | 8265                             | 8265                               | 8265                | 8265                  | 8265                       | 8265                       |
| Adjusted <i>R</i> <sup>2</sup> | 0.316               | 0.038                            | 0.236                              | 0.365               | 0.086                 | 0.028                      | 0.309                      |

This table reports coefficients from firm-panel regressions of specific governance risk incidents on the interaction between a firm's Big Three ownership and an indicator for years in and after 2013. The dependent variables are whether there are risk incidents on corruption (Column (1)), executive compensation (Column (2)), misleading communication (Column (3)), fraud (Column (4)), tax evasion (Column (5)), tax optimization (Column (6)), and anti competition (Column (7)). The sample includes firm-year observations from 2007 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

**TABLE AIX** Pre-Reform Big Three Ownership (2012) and Governance-Related Risk Incidents

| Dependent variable                            | (1)<br>Governance<br>issues | (2)<br>Corruption   | (3)<br>Executive<br>compensation | (4)<br>Misleading<br>communication | (5)<br>Fraud        | (6)<br>Tax<br>evasion | (7)<br>Tax<br>optimization | (8)<br>Anti<br>competition |
|-----------------------------------------------|-----------------------------|---------------------|----------------------------------|------------------------------------|---------------------|-----------------------|----------------------------|----------------------------|
| Big3 ownership <sup>2012</sup> × Post<br>2013 | 2.978***<br>(0.436)         | 0.767***<br>(0.211) | 0.023<br>(0.047)                 | 0.863***<br>(0.186)                | 2.156***<br>(0.361) | 0.309***<br>(0.116)   | 0.177***<br>(0.056)        | 1.572***<br>(0.378)        |
| Constant                                      | 0.120***<br>(0.024)         | 0.042***<br>(0.012) | 0.005<br>(0.003)                 | 0.034***<br>(0.011)                | 0.038**<br>(0.015)  | -0.001<br>(0.006)     | -0.003<br>(0.003)          | 0.058***<br>(0.020)        |
| Controls                                      | Yes                         | Yes                 | Yes                              | Yes                                | Yes                 | Yes                   | Yes                        | Yes                        |
| Firm FE                                       | Yes                         | Yes                 | Yes                              | Yes                                | Yes                 | Yes                   | Yes                        | Yes                        |
| Year FE                                       | Yes                         | Yes                 | Yes                              | Yes                                | Yes                 | Yes                   | Yes                        | Yes                        |
| N                                             | 8265                        | 8265                | 8265                             | 8265                               | 8265                | 8265                  | 8265                       | 8265                       |
| Adjusted R <sup>2</sup>                       | 0.409                       | 0.313               | 0.038                            | 0.230                              | 0.367               | 0.086                 | 0.028                      | 0.304                      |

This table reports coefficients from firm-panel regressions of governance risk incidents on the interaction between a firm's Big Three ownership in 2012 and an indicator for years in and after 2013. The dependent variables are whether there are governance risk incidents (Column (1)), risk incidents on corruption (Column (2)), executive compensation (Column (3)), misleading communication (Column (4)), fraud (Column (5)), tax evasion (Column (6)), tax optimization (Column (7)), and anti competition (Column (8)). The sample includes firm-year observations from 2007 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

**TABLE AX** Limited- and Medium-Reach Governance Risk Incidents

|                                                          | (1)                   | (2)                   | (3)                   | (4)                   | (5)                   | (6)                   | (7)                   | (8)                   | (9)                   | (10)                  |
|----------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Dependent variable                                       | Governanc<br>e issues | Governanc<br>e issues | Governanc<br>e issues | Governanc<br>e issues | Governanc<br>e issues | Governanc<br>e issues | Governanc<br>e issues | Governanc<br>e issues | Governanc<br>e issues | Governanc<br>e issues |
| Big3 ownership                                           | -0.174<br>(0.260)     | -0.167<br>(0.260)     | -0.072<br>(0.253)     | -0.006<br>(0.248)     | 0.006<br>(0.249)      | -0.167<br>(0.259)     | 0.002<br>(0.262)      | -0.173<br>(0.260)     | -0.135<br>(0.258)     | -0.040<br>(0.252)     |
| Big3 ownership×Post 2013                                 | 1.753***<br>(0.244)   | 1.728***<br>(0.247)   | 1.131***<br>(0.288)   | 0.943***<br>(0.260)   | 0.844***<br>(0.276)   | 1.702***<br>(0.247)   | 1.179***<br>(0.301)   | 1.736***<br>(0.254)   | 1.627***<br>(0.246)   | 1.049***<br>(0.288)   |
| Female zero <sup>2012</sup> ×Post 2013                   |                       | -0.027<br>(0.027)     |                       |                       |                       |                       |                       |                       |                       | -0.007<br>(0.028)     |
| Market capitalization <sup>2012</sup> ×Post2<br>013 2013 |                       |                       | 0.024***<br>(0.006)   |                       |                       |                       |                       |                       |                       | 0.024***<br>(0.006)   |
| Sales <sup>2012</sup> ×Post 2013                         |                       |                       |                       | 0.035***<br>(0.006)   |                       |                       |                       |                       |                       |                       |
| Assets <sup>2012</sup> ×Post 2013                        |                       |                       |                       |                       | 0.037***<br>(0.006)   |                       |                       |                       |                       |                       |
| MSCI <sup>2012</sup> ×Post 2013                          |                       |                       |                       |                       |                       | 0.059<br>(0.053)      |                       |                       |                       | 0.037<br>(0.052)      |
| S&P1200 <sup>2012</sup> ×Post 2013                       |                       |                       |                       |                       |                       |                       | 0.113***<br>(0.034)   |                       |                       |                       |
| Board size <sup>2012</sup> ×Post 2013                    |                       |                       |                       |                       |                       |                       |                       | 0.001<br>(0.002)      |                       | -0.003<br>(0.002)     |
| Board independence <sup>2012</sup> ×Post<br>2013         |                       |                       |                       |                       |                       |                       |                       |                       | 0.120***<br>(0.042)   | 0.088**<br>(0.042)    |
| Constant                                                 | 0.068***<br>(0.021)   | 0.082***<br>(0.024)   | -0.175***<br>(0.062)  | -0.321***<br>(0.066)  | -0.343***<br>(0.072)  | 0.068***<br>(0.021)   | 0.060***<br>(0.020)   | 0.065***<br>(0.022)   | 0.065***<br>(0.021)   | -0.165**<br>(0.068)   |
| Controls                                                 | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |
| Firm FE                                                  | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |
| Year FE                                                  | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |
| N                                                        | 8265                  | 8265                  | 8265                  | 8265                  | 8265                  | 8265                  | 8265                  | 8265                  | 8265                  | 8265                  |
| Adjusted R <sup>2</sup>                                  | 0.409                 | 0.409                 | 0.411                 | 0.414                 | 0.414                 | 0.409                 | 0.412                 | 0.409                 | 0.410                 | 0.412                 |

This table reports coefficients from firm-panel regressions of governance risk incidents on the interaction between a firm's Big Three ownership and an indicator for years in and after 2013. The dependent variable equals 1 if there are governance-related risk incidents with limited- and medium- reach information source, and 0 otherwise. Columns (2) include a control for no female director indicator in 2012 interacted with Post2013. Column (3) includes a control for log market cap in 2012 interacted with Post2013. Column (4) includes a control for log assets in 2012 interacted with Post2013. Column (5) includes a control for log sales in 2012 interacted with Post2013. Column (6) includes a control for MSCI Index inclusion in 2012 interacted with Post2013. Column (7) includes a control for S&P 1200 Index inclusion in 2012 interacted with Post2013. Column (8) includes a control for board size in 2012 interacted with Post2013. Column (9) includes a control for board independence in 2012 interacted with Post2013. Column (10) includes controls for the no-female-director indicator, log market capitalization, MSCI Index inclusion, board size, and board independence, with their 2012 values interacted with the Post-2013 indicator. The sample includes firm-year observations from 2007 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

## G. Causal Interpretation of Gender Diversity and Scrutiny

**TABLE AXI** Instrumental Variable Estimates: Female Director Ratio

| Dependent variable | (1)<br>Governance<br>issues | (2)<br>Corruption  | (3)<br>Executive<br>compensation | (4)<br>Misleading<br>communication | (5)<br>Fraud        | (6)<br>Tax<br>evasion | (7)<br>Tax<br>optimization | (8)<br>Anti<br>competition |
|--------------------|-----------------------------|--------------------|----------------------------------|------------------------------------|---------------------|-----------------------|----------------------------|----------------------------|
| Female ratio       | 3.765***<br>(0.848)         | 0.985**<br>(0.411) | 0.032<br>(0.051)                 | 1.265***<br>(0.282)                | 2.431***<br>(0.624) | 0.329**<br>(0.148)    | 0.209***<br>(0.080)        | 2.076***<br>(0.631)        |
| Big3 ownership     | -0.320<br>(0.454)           | 0.007<br>(0.261)   | -0.027<br>(0.020)                | -0.097<br>(0.108)                  | -0.253<br>(0.312)   | -0.026<br>(0.050)     | -0.004<br>(0.036)          | -0.105<br>(0.287)          |
| Controls           | Yes                         | Yes                | Yes                              | Yes                                | Yes                 | Yes                   | Yes                        | Yes                        |
| Firm FE            | Yes                         | Yes                | Yes                              | Yes                                | Yes                 | Yes                   | Yes                        | Yes                        |
| Year FE            | Yes                         | Yes                | Yes                              | Yes                                | Yes                 | Yes                   | Yes                        | Yes                        |
| <i>N</i>           | 8265                        | 8265               | 8265                             | 8265                               | 8265                | 8265                  | 8265                       | 8265                       |
| Kleibergen-Paap rk | 38.07                       | 38.07              | 38.07                            | 38.07                              | 38.07               | 38.07                 | 38.07                      | 38.07                      |
| Wald F statistic   |                             |                    |                                  |                                    |                     |                       |                            |                            |

This table reports coefficients from second-stage firm-panel regressions of governance risk incidents on the female ratio on the board using the interaction between a firm's Big Three ownership and an indicator for years in and after 2013 as an IV, firm and year fixed effects (FE). The dependent variables are whether there are governance risk incidents (Column (1)), risk incidents on corruption (Column (2)), executive compensation (Column (3)), misleading communication (Column (4)), fraud (Column (5)), tax evasion (Column (6)), tax optimization (Column (7)), and anti competition (Column (8)). The sample includes firm-year observations from 2007 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

**TABLE AXII** Instrumental Variable Estimates: Female Director Indicator

| Dependent variable | (1)<br>Governance<br>issues | (2)<br>Corruption  | (3)<br>Executive<br>compensation | (4)<br>Misleading<br>communication | (5)<br>Fraud        | (6)<br>Tax<br>evasion | (7)<br>Tax<br>optimization | (8)<br>Anti<br>competition |
|--------------------|-----------------------------|--------------------|----------------------------------|------------------------------------|---------------------|-----------------------|----------------------------|----------------------------|
| Female dummy       | 0.861***<br>(0.188)         | 0.225**<br>(0.092) | 0.007<br>(0.012)                 | 0.289***<br>(0.067)                | 0.556***<br>(0.139) | 0.075**<br>(0.033)    | 0.048***<br>(0.018)        | 0.475***<br>(0.151)        |
| Big3 ownership     | -0.301<br>(0.423)           | 0.012<br>(0.245)   | -0.027<br>(0.020)                | -0.091<br>(0.105)                  | -0.241<br>(0.305)   | -0.024<br>(0.047)     | -0.003<br>(0.036)          | -0.094<br>(0.299)          |
| Controls           | Yes                         | Yes                | Yes                              | Yes                                | Yes                 | Yes                   | Yes                        | Yes                        |
| Firm FE            | Yes                         | Yes                | Yes                              | Yes                                | Yes                 | Yes                   | Yes                        | Yes                        |
| Year FE            | Yes                         | Yes                | Yes                              | Yes                                | Yes                 | Yes                   | Yes                        | Yes                        |
| N                  | 8265                        | 8265               | 8265                             | 8265                               | 8265                | 8265                  | 8265                       | 8265                       |
| Kleibergen-Paap rk | 42.97                       | 42.97              | 42.97                            | 42.97                              | 42.97               | 42.97                 | 42.97                      | 42.97                      |
| Wald F statistic   |                             |                    |                                  |                                    |                     |                       |                            |                            |

This table reports coefficients from second-stage firm-panel regressions of governance risk incidents on the female indicator on the board using the interaction between a firm's Big Three ownership and an indicator for years in and after 2013 as an IV, firm and year fixed effects (FE). The dependent variables are whether there is governance risk incidents (Column (1)), risk incidents on corruption (Column (2)), executive compensation (Column (3)), misleading communication (Column (4)), fraud (Column (5)), tax evasion (Column (6)), tax optimization (Column (7)), and anti competition (Column (8)). The sample includes firm-year observations from 2007 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

## H. Alternative Explanation: Firm Performance and Governance Quality

**TABLE AXIII** Big Three Ownership and Firm Performance

| Dependent variable                               | (1)<br>ROE           | (2)<br>ROA           | (3)<br>CEO optimism  | (4)<br>Revenue growth | (5)<br>Employee growth |
|--------------------------------------------------|----------------------|----------------------|----------------------|-----------------------|------------------------|
| Big3 ownership                                   | -0.018<br>(0.051)    | 0.008<br>(0.020)     | 0.890<br>(2.177)     | -0.001<br>(0.066)     | -0.016<br>(0.058)      |
| Big3 ownership×Post 2013                         | 0.187***<br>(0.070)  | 0.053**<br>(0.023)   | -2.783<br>(2.085)    | 0.035<br>(0.076)      | 0.007<br>(0.067)       |
| Market capitalization <sup>2012</sup> ×Post 2013 | -0.007***<br>(0.002) | -0.003***<br>(0.001) | 0.194***<br>(0.055)  | -0.006***<br>(0.002)  | -0.004**<br>(0.002)    |
| Board independence <sup>2012</sup> ×Post 2013    | 0.007<br>(0.017)     | 0.004<br>(0.006)     | 0.558<br>(0.479)     | 0.023<br>(0.017)      | 0.025<br>(0.018)       |
| Constant                                         | 0.117***<br>(0.019)  | 0.046***<br>(0.007)  | -2.780***<br>(0.765) | 0.089***<br>(0.023)   | 0.071***<br>(0.020)    |
| Controls                                         | Yes                  | Yes                  | Yes                  | Yes                   | Yes                    |
| Firm FE                                          | Yes                  | Yes                  | Yes                  | Yes                   | Yes                    |
| Year FE                                          | Yes                  | Yes                  | Yes                  | Yes                   | Yes                    |
| <i>N</i>                                         | 8280                 | 8280                 | 6949                 | 8280                  | 7880                   |
| Adjusted <i>R</i> <sup>2</sup>                   | 0.180                | 0.278                | 0.127                | 0.442                 | 0.105                  |

This table reports coefficients from firm-panel regressions of firm performance and firm size growth on the interaction between a firm's Big Three ownership and an indicator for years in and after 2013. The dependent variables are ROE (Column (1)), ROA (Column (2)), forecast innovation (Column (3)), revenue growth (Column (4)), and employee growth (Column (5)). The sample includes firm-year observations from 2006 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table A1.



**TABLE AXIV** Big Three Ownership and Governance (Financial Reporting and External Audit Oversight)

| Dependent variable                               | (1)<br>Restatements | (2)<br>Revision number | (3)<br>Forecast gap | (4)<br>Earnings    | (5)<br>Audit fee     |
|--------------------------------------------------|---------------------|------------------------|---------------------|--------------------|----------------------|
| Big3 ownership                                   | 0.015<br>(0.054)    | 0.191<br>(0.656)       | 3.428***<br>(1.310) | -0.002<br>(0.435)  | 0.577<br>(0.395)     |
| Big3 ownership×Post 2013                         | 0.072<br>(0.077)    | 0.777<br>(0.823)       | -2.298<br>(1.497)   | -0.225<br>(0.442)  | 0.018<br>(0.447)     |
| Market capitalization <sup>2012</sup> ×Post 2013 | 0.003<br>(0.002)    | 0.039**<br>(0.020)     | 0.084**<br>(0.041)  | 0.012<br>(0.010)   | 0.081***<br>(0.011)  |
| Board independence <sup>2012</sup> ×Post 2013    | 0.001<br>(0.018)    | 0.016<br>(0.197)       | 0.264<br>(0.481)    | -0.145*<br>(0.078) | 0.133<br>(0.114)     |
| Constant                                         | -0.033<br>(0.024)   | 0.722***<br>(0.229)    | 0.117<br>(0.530)    | 0.157<br>(0.120)   | 12.519***<br>(0.124) |
| Controls                                         | Yes                 | Yes                    | Yes                 | Yes                | Yes                  |
| Firm FE                                          | Yes                 | Yes                    | Yes                 | Yes                | Yes                  |
| Year FE                                          | Yes                 | Yes                    | Yes                 | Yes                | Yes                  |
| <i>N</i>                                         | 8280                | 8073                   | 7476                | 7519               | 7608                 |
| Adjusted R <sup>2</sup>                          | 0.019               | 0.308                  | 0.254               | 0.066              | 0.927                |

This table reports coefficients from firm-panel regressions of governance (financial reporting and oversight) on the interaction between a firm's Big Three ownership and an indicator for years in and after 2013. The dependent variables are restatements (Column (1)), revision number (Column (2)), forecast gap (Column (3)), earnings surprise (Column (4)), and audit fees (Column (5)). The sample includes firm-year observations from 2006 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

**TABLE AXV** Incident Characteristics (Severity, Reach, and Novelty)

| Dependent variable       | (1)               | (2)                 | (3)              | (4)                 | (5)                 | (6)                  | (7)                 |
|--------------------------|-------------------|---------------------|------------------|---------------------|---------------------|----------------------|---------------------|
|                          | Severity          |                     | Reach            |                     | Novelty             |                      | Severe×reach mean   |
|                          | Dummy             | Mean                | Dummy            | Mean                | Dummy               | Mean                 |                     |
| Big3 ownership           | 0.146<br>(0.178)  | -0.508<br>(1.087)   | 0.328<br>(1.189) | 0.044<br>(0.891)    | -2.641**<br>(1.295) | -2.773***<br>(0.937) | -0.677<br>(1.808)   |
| Big3 ownership×Post 2013 | 0.239<br>(0.215)  | 0.439<br>(1.181)    | 0.314<br>(1.157) | 0.614<br>(0.858)    | 3.266***<br>(1.141) | 2.883***<br>(0.831)  | 1.851<br>(1.879)    |
| Constant                 | -0.064<br>(0.074) | 1.319***<br>(0.240) | 0.305<br>(0.225) | 1.863***<br>(0.184) | 0.535***<br>(0.162) | 1.362***<br>(0.156)  | 2.475***<br>(0.427) |
| Controls                 | Yes               | Yes                 | Yes              | Yes                 | Yes                 | Yes                  | Yes                 |
| Firm FE                  | Yes               | Yes                 | Yes              | Yes                 | Yes                 | Yes                  | Yes                 |
| Year FE                  | Yes               | Yes                 | Yes              | Yes                 | Yes                 | Yes                  | Yes                 |
| Constant                 | 590               | 590                 | 590              | 590                 | 585                 | 585                  | 585                 |
| Adjusted R <sup>2</sup>  | -0.040            | 0.196               | 0.214            | 0.119               | 0.058               | 0.223                | 0.062               |

This table reports coefficients from firm-panel regressions of severity, reach, and novelty of governance risk incidents on the interaction between a firm's Big Three ownership and an indicator for years in and after 2013. The dependent variables are severity (Columns (1) and (2)), reach (Columns (3) and (4)), novelty (Columns (5) and (6)), and severity multiplied by reach (Column (7)). The sample includes firm-year observations from 2007 through 2020. Standard errors, which are adjusted for clustering at the firm level, are reported in parentheses. All variables are defined in Online Appendix S1 Table AI.

## **I. Institutional Background: The “Nadeshiko Brand” Evaluation Process**

The Ministry of Economy, Trade and Industry (METI) selects outstanding companies through the Survey on Companies Encouraging Women’s Empowerment in the Workplace, which assesses firms’ efforts to promote women’s empowerment. Based on survey responses, METI evaluates companies and identifies those demonstrating innovative and effective practices

The survey focuses on three key areas:

- (1) initiatives to address gender pay gaps and related challenges.
- (2) efforts to implement diversity, equity, and inclusion (DEI) practices across their supply chains and partner networks.
- (3) measurable outcomes from programs addressing health-related issues for female employees.

An evaluation committee reviews the survey results and selects the top-performing firms, typically one to two companies per industry. On average, this corresponds to a 12% selection rate of all responding firms, which are designated as “Nadeshiko Brands.”

Source: Companies Selected under the FY2024 “Nadeshiko Brands” and “Next Nadeshiko: Companies Supporting Dual Careers and Co-parenting” (March, 2025)  
[https://www.meti.go.jp/english/press/2025/0324\\_001.html](https://www.meti.go.jp/english/press/2025/0324_001.html)

## J. Regression Models in Robustness Checks

1. Models to estimate using Big Three ownership in 2012:

$$GenderDiv_{it+1} = \beta_1 Big3\ ownership_i^{2012} \times Post2013_t + \beta_2 X_{it} + \alpha_i + \delta_t + \epsilon_{it} \quad (1)$$

$$Govern\ issues_{it+1} = \beta_1 Big3\ ownership_i^{2012} \times Post2013_t + \beta_2 X_{it} + \alpha_i + \delta_t + \epsilon_{it}$$

(2)

$Big3\ ownership_i^{2012}$  is the share of firm i's equity held by The Big Three in year 2012.

Our results are robust, using Big Three shareholdings in 2012 as treatment.

2. Models to estimate using Big Three ownership dummy:

$$GenderDiv_{it+1} = \beta_1 Big3\ high_i \times Post2013_t + \beta_2 X_{it} + \alpha_i + \delta_t + \epsilon_{it} \quad (3)$$

$Big3\ high_i$  is an indicator variable that equals 1 if firm i's Big Three holding is at or above the year-industry mean level during 2012 - 2020, and equals 0 if firm i's Big Three holding is below the year-industry mean level during 2012 - 2020.

3. Models for the Instrumental variable approach:

First stage:

$$GenderDiv_{it} = \beta_1 Big3\ ownership_{it} \times Post2013_t + \beta_2 Big3\ ownership_{it} + \beta_3 X_{it} + \alpha_i + \delta_t + \epsilon_{it} \quad (4)$$

Second stage:

$$Govern\ issues_{it+1} = \theta_1 \widehat{GenderDiv}_{it} + \theta_2 Big3\ ownership_{it} + \theta_3 X_{it} + \alpha_i + \delta_t + \epsilon_{it} \quad (5)$$